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Programming of cochlear implant accessories

Abstract

Pairing systems for pairing external devices to a cochlear implant system can comprise an external housing and an external pairing system. The external housing may comprise a first surface and one or more compartments, each configured to house an external device capable of wirelessly interfacing with an implantable cochlear implant system. The external pairing device may comprise a second surface and one or more corresponding near field communication devices. The near field communication devices may be arranged such that the first surface of the external housing can be aligned with the second surface of the external pairing device in such a way that each of the near field communication devices aligns with a corresponding compartment of the external housing. The external pairing device can provide communication between a programming device and external devices contained within compartments of the external housing via one or more corresponding near field communication devices.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2827041	12/1957	Pierson	N/A	N/A
4400590	12/1982	Michelson	N/A	N/A
4495384	12/1984	Scott et al.	N/A	N/A
4729366	12/1987	Schaefer	N/A	N/A
4850962	12/1988	Schaefer	N/A	N/A
4918745	12/1989	Hutchison	N/A	N/A
5540095	12/1995	Sherman et al.	N/A	N/A
5755742	12/1997	Schuelke et al.	N/A	N/A
5762583	12/1997	Adams et al.	N/A	N/A
6195585	12/2000	Karunasiri et al.	N/A	N/A
6212431	12/2000	Hahn et al.	N/A	N/A
6259951	12/2000	Kuzma et al.	N/A	N/A
6272382	12/2000	Faltys et al.	N/A	N/A
6308101	12/2000	Faltys et al.	N/A	N/A
6358281	12/2001	Berrang et al.	N/A	N/A
6391024	12/2001	Sun et al.	N/A	N/A
7225028	12/2006	Della Santina et al.	N/A	N/A
7319906	12/2007	Kuzma et al.	N/A	N/A
7524278	12/2008	Madsen et al.	N/A	N/A
7729775	12/2009	Saoji et al.	N/A	N/A
8554329	12/2012	Mann et al.	N/A	N/A
8626308	12/2013	Meskens	N/A	N/A
8655449	12/2013	Haller et al.	N/A	N/A
8660658	12/2013	Walsh et al.	N/A	N/A
8954158	12/2014	Smith	N/A	N/A
9061140	12/2014	Shi et al.	N/A	N/A
9205272	12/2014	Suaning et al.	N/A	N/A
9504076	12/2015	El-Hoiydi	N/A	H04W 12/50
9539430	12/2016	Mishra et al.	N/A	N/A
9693155	12/2016	Meister et al.	N/A	N/A
9716952	12/2016	Mauger	N/A	N/A

9999770	12/2017	Walraevens et al.	N/A	N/A
11324958	12/2021	Anderson et al.	N/A	N/A
2002/0015506	12/2001	Aceti et al.	N/A	N/A
2002/0039425	12/2001	Burnett et al.	N/A	N/A
2002/0099421	12/2001	Goldsmith et al.	N/A	N/A
2004/0230254	12/2003	Harrison et al.	N/A	N/A
2005/0033384	12/2004	Sacha	N/A	N/A
2005/0197677	12/2004	Stevenson	N/A	N/A
2006/0122664	12/2005	Sacha et al.	N/A	N/A
2006/0183965	12/2005	Kasic et al.	N/A	N/A
2008/0195179	12/2007	Quick	N/A	N/A
2008/0300658	12/2007	Meskens	N/A	N/A
2009/0018616	12/2008	Quick et al.	N/A	N/A
2009/0082831	12/2008	Paul et al.	N/A	N/A
2009/0171421	12/2008	Atalar et al.	N/A	N/A
2009/0187233	12/2008	Stracener	N/A	N/A
2009/0192565	12/2008	Lee et al.	N/A	N/A
2010/0010582	12/2009	Carbunaru et al.	N/A	N/A
2010/0030012	12/2009	Meskens	N/A	N/A
2010/0042183	12/2009	Beck	N/A	N/A
2010/0317913	12/2009	Conn et al.	N/A	N/A
2011/0082521	12/2010	Botros et al.	N/A	N/A
2011/0098785	12/2010	Mishra	N/A	N/A
2011/0116669	12/2010	Karunasiri	N/A	N/A
2011/0137180	12/2010	Johnson et al.	N/A	N/A
2011/0144719	12/2010	Perkins et al.	N/A	N/A
2011/0160808	12/2010	Lyden et al.	N/A	N/A
2011/0280426	12/2010	Bachler	N/A	N/A
2011/0295331	12/2010	Wells et al.	N/A	N/A
2012/0063610	12/2011	Kaulberg et al.	N/A	N/A
2012/0215285	12/2011	Tahmasian et al.	N/A	N/A
2012/0277835	12/2011	Della Santina et al.	N/A	N/A
2013/0018216	12/2012	Beckerle et al.	N/A	N/A
2013/0023953	12/2012	Van Den Honert	N/A	N/A
2013/0193914	12/2012	Gaddam et al.	N/A	N/A
2013/0197613	12/2012	Kelly et al.	N/A	N/A
2013/0223664	12/2012	Meskens et al.	N/A	N/A
2013/0238055	12/2012	Marnfeldt et al.	N/A	N/A
2013/0268025	12/2012	Ranu	N/A	N/A
2013/0278226	12/2012	Cong et al.	N/A	N/A
2013/0317584	12/2012	Stevenson et al.	N/A	N/A
2014/0058482	12/2013	Gupta et al.	N/A	N/A
2014/0070761	12/2013	Labbe et al.	N/A	N/A
2014/0155947	12/2013	Kroll et al.	N/A	N/A
2014/0247954	12/2013	Hall et al.	N/A	N/A
2014/0270211	12/2013	Solum et al.	N/A	N/A
2014/0275730	12/2013	Lievens et al.	N/A	N/A
2014/0309712	12/2013	Masaki et al.	N/A	N/A
2014/0350652	12/2013	Suwito	N/A	N/A
2015/0125012	12/2014	Sabin	N/A	N/A

2015/0174416	12/2014	Angara et al.	N/A	N/A
2015/0224312	12/2014	Platz et al.	N/A	N/A
2015/0256945	12/2014	Mazanec	N/A	N/A
2015/0374988	12/2014	Laudanski	N/A	N/A
2015/0375003	12/2014	Meskens	N/A	N/A
2016/0050500	12/2015	Liao et al.	N/A	N/A
2016/0227333	12/2015	Babico	N/A	N/A
2017/0043162	12/2016	Lopez-Poveda	N/A	N/A
2017/0077938	12/2016	Heubi	N/A	N/A
2017/0094396	12/2016	Chandramohan	N/A	H04R 1/345
2017/0161449	12/2016	Meskens	N/A	N/A
2017/0259072	12/2016	Newham et al.	N/A	N/A
2017/0360364	12/2016	Heasman et al.	N/A	N/A
2018/0028811	12/2017	Van Gerwen et al.	N/A	N/A
2018/0028827	12/2017	Schilling et al.	N/A	N/A
2018/0041848	12/2017	Nielsen et al.	N/A	N/A
2018/0050197	12/2017	Mazanec et al.	N/A	N/A
2018/0050198	12/2017	Mazanec et al.	N/A	N/A
2018/0050203	12/2017	Mazanec et al.	N/A	N/A
2018/0059870	12/2017	Krah	N/A	N/A
2018/0264269	12/2017	Meadows	N/A	N/A
2018/0317027	12/2017	Bolner et al.	N/A	N/A
2018/0333577	12/2017	Nygaard et al.	N/A	N/A
2018/0361151	12/2017	Ridler et al.	N/A	N/A
2019/0045308	12/2018	Chen et al.	N/A	N/A
2019/0046116	12/2018	Shah et al.	N/A	N/A
2019/0190296	12/2018	Paralikar et al.	N/A	N/A
2019/0217101	12/2018	Shi et al.	N/A	N/A
2019/0231203	12/2018	Harczos	N/A	N/A
2019/0344073	12/2018	Baker et al.	N/A	N/A
2019/0358450	12/2018	Lo et al.	N/A	N/A
2020/0054877	12/2019	Calixto et al.	N/A	N/A
2020/0238075	12/2019	Mazanec et al.	N/A	N/A
2020/0269034	12/2019	Mazanec et al.	N/A	N/A
2020/0269035	12/2019	Mazanec et al.	N/A	N/A
2020/0269047	12/2019	Mazanec et al.	N/A	N/A
2020/0269048	12/2019	Mazanec et al.	N/A	N/A
2020/0269057	12/2019	Mazanec et al.	N/A	N/A
2020/0269058	12/2019	Mazanec et al.	N/A	N/A
2021/0084417	12/2020	Bagazov	N/A	H04R 25/65
2021/0121707	12/2020	Fried et al.	N/A	N/A
2021/0135704	12/2020	El-Hoiydi et al.	N/A	N/A
2021/0187293	12/2020	Friedling	N/A	N/A
2021/0361194	12/2020	Arab et al.	N/A	N/A
2022/0203104	12/2021	Hernandez et al.	N/A	N/A
2022/0339445	12/2021	Litvak et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
104394930	12/2014	CN	N/A

110086237	12/2018	CN	N/A
4419070	12/1993	DE	N/A
60107062	12/2003	DE	N/A
102013214049	12/2014	DE	N/A
1043914	12/1999	EP	N/A
1683544	12/2009	EP	N/A
2884766	12/2017	EP	N/A
3120579	12/2019	EP	N/A
2016024111	12/2015	JP	N/A
201142830	12/2010	TW	N/A
2007137032	12/2006	WO	N/A
2010056768	12/2009	WO	N/A
2014037888	12/2013	WO	N/A
2014054215	12/2013	WO	N/A
2015077773	12/2014	WO	N/A
2016122606	12/2015	WO	N/A
2017100866	12/2016	WO	N/A
2018035329	12/2017	WO	N/A
2018144732	12/2017	WO	N/A
2020172500	12/2019	WO	N/A

OTHER PUBLICATIONS

PCT International Search Report and Written Opinion dated Jan. 5, 2022 for related Intl. App. No. PCT/US2021/047855, 12 pages. cited by applicant

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 17/006,467, filed Aug. 28, 2020, entitled “PROGRAMMING OF COCHLEAR IMPLANT ACCESSORIES,” the entire contents of which are incorporated herein by reference.

BACKGROUND

(1) Cochlear implant systems are systems which may be at least partially implanted surgically into and around the cochlea, the hearing organ of the ear, to provide improved hearing to a patient. Cochlear implant systems may also be in communication with various devices internal to the wearer as well as one or more external devices.

(2) External devices may comprise a variety of components that can be used to interact with the cochlear implant system or in tandem with the cochlear implant system to provide a better hearing experience for the patient. However in some cases, external devices may need to be paired with the cochlear implant system prior to them communicating with the cochlear implant system. Manually pairing the various external devices with a wearer's implant system may prove to be costly both monetarily as well as in time. For example, an audiologist may need to electrically charge and manually pair various devices with a wearer's cochlear implant system one-by-one. Charging each device enough to turn on and pair each device to the wearer's cochlear implant system can take an

undesirably long time, and additional device to be paired can compound the delay. Furthermore, manually pairing each external device leaves room for human error, such as forgetting to charge one or more external devices or forgetting to pair one or more external devices before sending such a device home with the wearer. This may subsequently require a second appointment to fix any errors or pair missed devices.

SUMMARY

(3) Some aspects of the disclosure are generally directed toward a pairing system for pairing one or more external devices to a cochlear implant system. In some examples, the pairing system may comprise an external housing, such as a box, a briefcase, or the like. The external housing may have a first surface as well as a plurality of compartments arranged in a first configuration such that each of the plurality of compartments has a unique position within the external housing relative to the first surface. Each of the plurality of compartments may be additionally configured to house an external device capable of wirelessly interfacing with a cochlear implant system, such as a fully implantable cochlear implant system. In some embodiments, each of the plurality of compartments may be shaped as to receive a unique corresponding external device however alternative shapes may be used.

(4) Additionally or alternatively, the pairing device may include an external pairing device, such as a mat, a tabletop, or the like. The external pairing device may comprise a second surface as well as a plurality of near field communication devices. In some embodiments, the plurality of near field communication devices may be arranged in a configuration corresponding to the first configuration and/or arranged in the first configuration relative to the second surface such that the first surface of the external housing can be aligned with the second surface of the external pairing device in such a way that each of the plurality of near field communication devices aligns with a corresponding one of the plurality of compartments of the external housing. In some embodiments, the plurality of near field communication devices may be located beneath the second surface. The external pairing device may be configured to provide communication between a programming device and one or more external devices, each contained with a different compartment of the external housing, via one or more corresponding near field communication devices of the external pairing device.

(5) In some embodiments, the external pairing device may be configured to electrically charge one or more external devices, each contained in a corresponding compartment of the external housing, via a corresponding one or more near field communication device of the external pairing device when the external housing is positioned proximate the external pairing device and each of the plurality of near field communication devices aligns with a corresponding one of the plurality of compartments of the external housing. Additionally or alternatively, each near field communication device comprises a coil configured to facilitate communication with and charging of an external device within a corresponding compartment and having a corresponding coil.

(6) In some embodiments, systems may comprise a cochlear implant system, a programming device, and one or more external devices. The cochlear implant system may comprise a cochlear electrode, a stimulator, a signal processor, and an implantable battery and/or communication module. In some embodiments, the programming device is configured to communicate with each of the one or more external devices via corresponding near field communication devices of the external pairing device and provide information to each of the one or more external devices to enable communication between each external device and the cochlear implant system. Additionally or alternatively, the programming device may be configured to provide information to each of the one or more external devices to enable communication between each external device and the implantable battery and/or communication module of the cochlear implant system.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 shows a schematic illustration of a fully implantable cochlear implant system.
- (2) FIG. 2 shows an embodiment of a fully implantable cochlear implant.
- (3) FIG. 3 illustrates an embodiment of an exemplary middle ear sensor for use in conjunction with anatomical features of a patient
- (4) FIG. 4 is a schematic diagram illustrating an exemplary implantable system including an acoustic stimulator.
- (5) FIG. 5A is a high level electrical schematic showing communication between the implantable battery and/or communication module and the signal processor.
- (6) FIG. 5B illustrates an exemplary schematic diagram illustrating a cochlear electrode having a plurality of contact electrodes and fixedly or detachably connected to an electrical stimulator.
- (7) FIG. 6A shows a high level schematic diagram illustrating an exemplary communication configuration between an implantable battery and/or communication module, a signal processor, and a stimulator in an exemplary cochlear implant system.
- (8) FIG. 6B is a schematic diagram illustrating exemplary electrical communication between an implantable battery and/or communication module and a signal processor in a cochlear implant system according to some embodiments.
- (9) FIG. 7A is an alternative high-level schematic diagram illustrating an exemplary communication configuration between an implantable battery and/or communication module, a signal processor, and a stimulator.
- (10) FIG. 7B is an alternative schematic diagram illustrating exemplary electrical communication between an implantable battery and/or communication module and a signal processor in a cochlear implant system similar to that shown in FIG. 7A.
- (11) FIG. 7C is another alternative schematic diagram illustrating exemplary electrical communication between an implantable battery and/or communication module and a signal processor in a cochlear implant system similar to that shown in FIG. 7A.
- (12) FIG. 7D is high-level schematic diagram illustrating exemplary electrical communication between an implantable battery and/or communication module and a signal processor in a cochlear implant system similar to that shown in FIG. 7A.
- (13) FIG. 8 is a process flow diagram illustrating an exemplary process for establishing a preferred transfer function for a patient.
- (14) FIG. 9 is a schematic diagram illustrating possible communication between a variety of system components according to some embodiments of a fully implantable system.
- (15) FIG. 10 is an exemplary illustration of an external housing and an external pairing device for pairing one or more external devices with an implantable cochlear implant system.
- (16) FIG. 11 is an exemplary illustration of an external device establishing a connection with an implantable cochlear implant system and a programming device.
- (17) FIG. 12 provides an exemplary method of establishing a connection between an external device and an implantable cochlear implant system.
- (18) FIG. 13 shows a process flow diagram showing an exemplary method for pairing another device with an implanted system using a paired charger.
- (19) FIG. 14 is a chart showing the various parameters that are adjustable by each of a variety of external devices.

DETAILED DESCRIPTION

- (20) FIG. 1 shows a schematic illustration of a fully implantable cochlear implant system. The system of FIG. 1 includes a middle ear sensor **110** in communication with a signal processor **120**. The middle ear sensor **110** can be configured to detect incoming sound waves, for example, using the ear structure of a patient. The signal processor **120** can be configured to receive a signal from the middle ear sensor **110** and produce an output signal based thereon. For example, the signal

processor **120** can be programmed with instructions to output a certain signal based on a received signal. In some embodiments, the output of the signal processor **120** can be calculated using an equation based on received input signals. Alternatively, in some embodiments, the output of the signal processor **120** can be based on a lookup table or other programmed (e.g., in memory) correspondence between the input signal from the middle ear sensor **110** and the output signal. While not necessarily based explicitly on a function, the relationship between the input to the signal processor **120** (e.g., from the middle ear sensor **110**) and the output of the signal processor **120** is referred to as the transfer function of the signal processor **120**.

(21) The system of FIG. **1** further includes a cochlear electrode **116** implanted into the cochlear tissues of a patient. The cochlear electrode **116** is in electrical communication with an electrical stimulator **130**, which can be configured to provide electrical signals to the cochlear electrode **116** in response to input signals received by the electrical stimulator **130**. In some examples, the cochlear electrode **116** is fixedly attached to the electrical stimulator **130**. In other examples, the cochlear electrode **116** is removably attached to the electrical stimulator **130**. As shown, the electrical stimulator **130** is in communication with the signal processor **120**. In some embodiments, the electrical stimulator **130** provides electrical signals to the cochlear electrode **116** based on output signals from the signal processor **120**.

(22) In various embodiments, the cochlear electrode **116** can include any number of contact electrodes in electrical contact with different parts of the cochlear tissue. In such embodiments, the electrical stimulator **130** can be configured to provide electrical signals to any number of such contact electrodes to stimulate the cochlear tissue. For example, in some embodiments, the electrical stimulator **130** is configured to activate different contact electrodes or combinations of contact electrodes of the cochlear electrode **116** in response to different input signals received from the signal processor **120**. This can help the patient differentiate between different input signals.

(23) During exemplary operation, the middle ear sensor **110** detects audio signals, for example, using features of the patient's ear anatomy as described elsewhere herein and in U.S. Patent Publication No. 2013/0018216, which is hereby incorporated by reference in its entirety. The signal processor **120** can receive such signals from the middle ear sensor **110** and produce an output to the electrical stimulator **130** based on the transfer function of the signal processor **120**. The electrical stimulator **130** can then stimulate one or more contact electrodes of the cochlear electrode **116** based on the received signals from the signal processor **120**.

(24) Referring to FIG. **2**, an embodiment of a fully implantable cochlear implant is shown. The device in this embodiment includes a processor **220** (e.g., signal processor), a sensor **210**, a first lead **270** connecting the sensor **210** to the processor **220**, and a combination lead **280** attached to the processor **220**, wherein combination lead **280** contains both a ground electrode **217** and a cochlear electrode **216**. The illustrated processor **220** includes a housing **202**, a coil **208**, first female receptacle **271** and second female receptacle **281** for insertion of the leads **270** and **280**, respectively.

(25) In some embodiments, coil **208** can receive power and/or data from an external device, for instance, including a transmission coil (not shown). Some such examples are described in U.S. Patent Publication No. 2013/0018216, which is incorporated by reference. In other examples, processor **220** is configured to receive power and/or data from other sources, such as an implantable battery and/or communication module as shown in FIG. **1**. Such battery and/or communication module can be implanted, for example, into the pectoral region of the patient in order to provide adequate room for larger equipment (e.g., a relatively large battery) for prolonged operation (e.g., longer battery life). Additionally, in the event a battery needs eventual replacement, a replacement procedure in the patient's pectoral region can be performed several times without certain vascularization issues that can arise near the location of the cochlear implant. For example, in some cases, repeated procedures (e.g., battery replacement) near the cochlear implant can result in a decreased ability for the skin in the region to heal after a procedure. Placing a replaceable

component such as a battery in the pectoral region can facilitate replacement procedures with reduced risk for such issues.

(26) FIG. 3 illustrates embodiments of an exemplary middle ear sensor for use in conjunction with anatomical features of a patient. Referring to FIG. 3, an embodiment of the sensor **310** of a fully implantable cochlear implant is shown. Also shown are portions of the subject's anatomy, which includes, if the subject is anatomically normal, at least the malleus **322**, incus **324**, and stapes **326** of the middle ear **328**, and the cochlea **348**, oval window **346**, and round window **344** of the inner ear **342**. Here, the sensor **310** is touching the incus **324**. Further, although not shown in a drawing, the sensor **310** may be in operative contact with the tympanic membrane or the stapes, or any combination of the tympanic membrane, malleus **322**, incus **324**, or stapes **326**.

(27) FIG. 3 illustrates an exemplary middle ear sensor for use with systems described herein. However, other middle ear sensors can be used, such as sensors using microphones or other sensors capable of receiving an input corresponding to detected sound and outputting a corresponding signal to the signal processor. Additionally or alternatively, systems can include other sensors configured to output a signal representative of sound received at or near a user's ear, such as a microphone or other acoustic pickup located in the user's outer ear or implanted under the user's skin. Such devices may function as an input source, for example, to the signal processor such that the signal processor receives an input signal from the input source and generates and output one or more stimulation signals according to the received input signal and the signal processor transfer function.

(28) Referring back to FIG. 1, the signal processor **120** is shown as being in communication with the middle ear sensor **110**, the electrical stimulator **130**, and the implantable battery and/or communication module **140**. As described elsewhere herein, the signal processor **120** can receive input signals from the middle ear sensor **110** and/or other input source(s) and output signals to the electrical stimulator **130** for stimulating the cochlear electrode **116**. The signal processor **120** can receive data (e.g., processing data establishing or updating the transfer function of the signal processor **120**) and/or power from the implantable battery and/or communication module **140**.

(29) In some embodiments, the implantable battery and/or communication module **140** can communicate with external devices, such as a programmer **100** and/or a battery charger **102**. The battery charger **102** can wirelessly charge the battery in the implantable battery and/or communication module **140** when brought into proximity with the implantable battery and/or communication module **140** in the pectoral region of the patient. Such charging can be accomplished, for example, using inductive charging. The programmer **100** can be configured to wirelessly communicate with the implantable battery and/or communication module **140** via any appropriate wireless communication technology, such as Bluetooth, Wi-Fi, and the like. In some examples, the programmer **100** can be used to update the system firmware and/or software. In an exemplary operation, the programmer **100** can be used to communicate an updated signal processor **120** transfer function to the implantable battery and/or communication module **140**. In various embodiments, the programmer **100** and charger **102** can be separate devices or can be integrated into a single device.

(30) In the illustrated example of FIG. 1, the signal processor **120** is connected to the middle ear sensor **110** via lead **170**. In some embodiments, lead **170** can provide communication between the signal processor **120** and the middle ear sensor **110**. In some embodiments, lead **170** can include a plurality of isolated conductors providing a plurality of communication channels between the middle ear sensor **110** and the signal processor **120**. The lead **170** can include a coating such as an electrically insulating sheath to minimize any conduction of electrical signals to the body of the patient. In various embodiments, one or more communication leads can be detachable such that communication between two components can be disconnected in order to electrically and/or mechanically separate such components. For instance, in some embodiments, lead **170** includes a detachable connector **171**. Detachable connector **171** can facilitate decoupling of the signal

processor **120** and middle ear sensor **110**. For example, with reference to FIG. **1**, in some embodiments, lead **170** can include a first lead extending from the middle ear sensor **110** having one of a male (e.g., **672**) or a female (e.g., **673**) connector and a second lead extending from the signal processor **120** having the other of the male or female connector. The first and second leads can be connected at detachable connector **171** in order to facilitate communication between the middle ear sensor **110** and the signal processor **120**.

(31) In other examples, a part of the detachable connector **171** can be integrated into one of the middle ear sensor **110** and the signal processor **120**. For example, in an exemplary embodiment, the signal processor **120** can include a female connector integrated into a housing of the signal processor **120**. Lead **170** can extend fully from the middle ear sensor **110** and terminate at a corresponding male connector for inserting into the female connector of the signal processor **120**. In still further embodiments, a lead (e.g., **170**) can include connectors on each end configured to detachably connect with connectors integrated into each of the components in communication. For example, lead **170** can include two male connectors, two female connectors, or one male and one female connector for detachably connecting with corresponding connectors integral to the middle ear sensor **110** and the signal processor **120**. Thus, lead **170** may include two or more detachable connectors.

(32) Similar communication configurations can be established for detachable connector **181** of lead **180** facilitating communication between the signal processor **120** and the stimulator **130** and for detachable connector **191** of lead **190** facilitating communication between the signal processor **120** and the implantable battery and/or communication module **140**. Leads (**170**, **180**, **190**) can include pairs of leads having corresponding connectors extending from each piece of communicating equipment, or connectors can be built in to any one or more communicating components.

(33) In such configurations, each of the electrical stimulator **130**, signal processor **120**, middle ear sensor **110**, and battery and/or communication module can each be enclosed in a housing, such as a hermetically sealed housing comprising biocompatible materials. Such components can include feedthroughs providing communication to internal components enclosed in the housing.

Feedthroughs can provide electrical communication to the component via leads extending from the housing and/or connectors integrated into the components.

(34) In a module configuration such as that shown in FIG. **1**, various components can be accessed (e.g., for upgrades, repair, replacement, etc.) individually from other components. For example, as signal processor **120** technology improves (e.g., improvements in size, processing speed, power consumption, etc.), the signal processor **120** implanted as part of the system can be removed and replaced independently of other components. In an exemplary procedure, an implanted signal processor **120** can be disconnected from the electrical stimulator **130** by disconnecting detachable connector **181**, from the middle ear sensor **110** by disconnecting detachable connector **171**, and from the implantable battery and/or communication module **140** by disconnecting detachable connector **191**. Thus, the signal processor **120** can be removed from the patient while other components such as the electrical stimulator **130**, cochlear electrode **116**, middle ear sensor **110**, and battery and/or communication module can remain in place in the patient.

(35) After the old signal processor is removed, a new signal processor can be connected to the electrical stimulator **130**, middle ear sensor **110**, and implantable battery and/or communication module **140** via detachable connectors **181**, **171**, and **191**, respectively. Thus, the signal processor (e.g., **120**) can be replaced, repaired, upgraded, or any combination thereof, without affecting the other system components. This can reduce, among other things, the risk, complexity, duration, and recovery time of such a procedure. In particular, the cochlear electrode **116** can be left in place in the patient's cochlea while other system components can be adjusted, reducing trauma to the patient's cochlear tissue.

(36) Such modularity of system components can be particularly advantageous when replacing a signal processor **120**, such as described above. Processor technology continues to improve and will

likely continue to markedly improve in the future, making the signal processor **120** a likely candidate for significant upgrades and/or replacement during the patient's lifetime. Additionally, in embodiments such as the embodiment shown in FIG. **1**, the signal processor **120** communicates with many system components. For example, as shown, the signal processor **120** is in communication with each of the electrical stimulator **130**, the middle ear sensor **110**, and the implantable battery and/or communication module **140**. Detachably connecting such components with the signal processor **120** (e.g., via detachable connectors **181**, **171**, and **191**) enables replacement of the signal processor **120** without disturbing any other components. Thus, in the event of an available signal processor **120** upgrade and/or a failure of the signal processor **120**, the signal processor **120** can be disconnected from other system components and removed.

(37) While many advantages exist for a replaceable signal processor **120**, the modularity of other system components can be similarly advantageous, for example, for upgrading any system component. Similarly, if a system component (e.g., the middle ear sensor **110**) should fail, the component can be disconnected from the rest of the system (e.g., via detachable connector **171**) and replaced without disturbing the remaining system components. In another example, even a rechargeable battery included in the implantable battery and/or communication module **140** may eventually wear out and need replacement. The implantable battery and/or communication module **140** can be replaced or accessed (e.g., for replacing the battery) without disturbing other system components. Further, as discussed elsewhere herein, when the implantable battery and/or communication module **140** is implanted in the pectoral region of the patient, such as in the illustrated example, such a procedure can leave the patient's head untouched, eliminating unnecessarily frequent access beneath the skin.

(38) While various components are described herein as being detachable, in various embodiments, one or more components configured to communicate with one another can be integrated into a single housing. For example, in some embodiments, signal processor **120** can be integrally formed with the stimulator **130** and cochlear electrode **116**. For example, in an exemplary embodiment, processing and stimulation circuitry of a signal processor **120** and stimulator **130** can be integrally formed as a single unit in a housing coupled to a cochlear electrode. Cochlear electrode and the signal processor/stimulator can be implanted during an initial procedure and operate as a single unit.

(39) In some embodiments, while the integral signal processor/stimulator/cochlear electrode component does not get removed from a patient due to potential damage to the cochlear tissue into which the cochlear electrode is implanted, system upgrades are still possible. For example, in some embodiments, a module signal processor may be implanted alongside the integral signal processor/stimulator component and communicate therewith. In some such examples, the integral signal processor may include a built-in bypass to allow a later-implanted signal processor to interface directly with the stimulator. Additionally or alternatively, the modular signal processor can communicate with the integral signal processor, which may be programmed with a unity transfer function. Thus, in some such embodiments, signals from the modular signal processor may be essentially passed through the integral signal processor unchanged so that the modular signal processor effectively controls action of the integral stimulator. Thus, in various embodiments, hardware and/or software solutions exist for upgrading an integrally attached signal processor that may be difficult or dangerous to remove.

(40) While often described herein as using an electrical stimulator to stimulate the patient's cochlear tissue via a cochlear electrode, in some examples, the system can additionally or alternatively include an acoustic stimulator. An acoustic stimulator can include, for example, a transducer (e.g., a piezoelectric transducer) configured to provide mechanical stimulation to the patient's ear structure. In an exemplary embodiment, the acoustic stimulator can be configured to stimulate one or more portions of the patient's ossicular chain via amplified vibrations. Acoustic stimulators can include any appropriate acoustic stimulators, such as those found in the

ESTEEM™ implant (Envoy Medical Corp., St. Paul, Minn.) or as described in U.S. Pat. Nos. 4,729,366, 4,850,962, and 7,524,278, and U.S. Patent Publication No. 20100042183, each of which is incorporated herein by reference in its entirety.

(41) FIG. 4 is a schematic diagram illustrating an exemplary implantable system including an acoustic stimulator. The acoustic stimulator can be implanted proximate the patient's ossicular chain and can be in communication with a signal processor via lead **194** and detachable connector **195**. The signal processor can behave as described elsewhere herein and can be configured to cause acoustic stimulation of the ossicular chain via the acoustic stimulator in response to input signals from the middle ear sensor according to a transfer function of the signal processor.

(42) The acoustic stimulator of FIG. 4 can be used similarly to the electrical stimulator as described elsewhere herein. For instance, an acoustic stimulator can be mechanically coupled to a patient's ossicular chain upon implanting the system and coupled to the signal processor via lead **194** and detachable connector **195**. Similarly to systems described elsewhere herein with respect to the electrical stimulator, if the signal processor requires replacement or repair, the signal processor can be disconnected from the acoustic stimulator (via detachable connector **195**) so that the signal processor can be removed without disturbing the acoustic stimulator.

(43) Some systems can include a hybrid system comprising both an electrical stimulator and an acoustic stimulator in communication with the signal processor. In some such examples, the signal processor can be configured to stimulate electrically and/or acoustically according to the transfer function of the signal processor. In some examples, the type of stimulation used can depend on the input signal received by the signal processor. For instance, in an exemplary embodiment, the frequency content of the input signal to the signal processor can dictate the type of stimulation. In some cases, frequencies below a threshold frequency could be represented using one of electrical and acoustic stimulation while frequencies above the threshold frequency could be represented using the other of electrical and acoustic stimulation. Such a threshold frequency could be adjustable based on the hearing profile of the patient. Using a limited range of frequencies can reduce the number of frequency domains, and thus the number of contact electrodes, on the cochlear electrode. In other examples, rather than a single threshold frequency defining which frequencies are stimulated electrically and acoustically, various frequencies can be stimulated both electrically and acoustically. In some such examples, the relative amount of electrical and acoustic stimulation can be frequency-dependent. As described elsewhere herein, the signal processor transfer function can be updated to meet the needs of the patient, including the electrical and acoustic stimulation profiles.

(44) With further reference to FIGS. 1 and 4, in some examples, a system can include a shut-off controller **104**, which can be configured to wirelessly stop an electrical stimulator **130** from stimulating the patient's cochlear tissue and/or an acoustic stimulator **150** from stimulating the patient's ossicular chain. For example, if the system is malfunctioning or an uncomfortably loud input sound causes an undesirable level of stimulation, the user may use the shut-off controller **104** to cease stimulation from the stimulator **130**. The shut-off controller **104** can be embodied in a variety of ways. For example, in some embodiments, the shut-off controller **104** can be integrated into other external devices, such as the programmer **100**. In some such examples, the programmer **100** includes a user interface by which a user can select an emergency shut-off feature to cease stimulation. Additionally or alternatively, the shut-off controller **104** can be embodied as a separate component. This can be useful in situations in which the patient may not have immediate access to the programmer **100**. For example, the shut-off controller **104** can be implemented as a wearable component that the patient can wear at all or most times, such as a ring, bracelet, necklace, or the like.

(45) The shut-off controller **104** can communicate with the system in order to stop stimulation in a variety of ways. In some examples, the shut-off controller **104** comprises a magnet that is detectable by a sensor (e.g., a Hall-Effect sensor) implanted in the patient, such as in the processor

and/or the implantable battery and/or communication module **140**. In some such embodiments, when the magnet is brought sufficiently close to the sensor, the system can stop stimulation of the cochlear tissue or ossicular chain.

(46) After the shut-off controller **104** is used to disable stimulation, stimulation can be re-enabled in one or more of a variety of ways. For example, in some embodiments, stimulation is re-enabled after a predetermined amount of time after it had been disabled. In other examples, the shut-off controller **104** can be used to re-enable stimulation. In some such examples, the patient brings the shut-off controller **104** within a first distance of a sensor (e.g., a magnetic sensor) to disable stimulation, and then removes the shut-off controller **104**. Subsequently, once the patient brings the shut-off controller **104** within a second distance of the sensor, stimulation can be re-enabled. In various embodiments, the first distance can be less than the second distance, equal to the second distance, or greater than the second distance. In still further embodiments, another device such as a separate turn-on controller (not shown) or the programmer **100** can be used to re-enable stimulation. Any combination of such re-enabling of stimulation can be used, such as alternatively using either the programmer **100** or the shut-off controller **104** to enable stimulation or combining a minimum “off” time before any other methods can be used to re-enable stimulation.

(47) In some embodiments, rather than entirely disable stimulation, other actions can be taken, such as reducing the magnitude of stimulation. For example, in some embodiments, the shut-off sensor can be used to reduce the signal output by a predetermined amount (e.g., absolute amount, percentage, etc.). In other examples, the shut-off sensor can affect the transfer function of the signal processor to reduce the magnitude of stimulation in a customized way, such as according to frequency or other parameter of an input signal (e.g., from the middle ear sensor).

(48) With reference back to FIG. **1**, as described elsewhere herein, the implantable battery and/or communication module can be used to provide power and/or data (e.g., processing instructions) to other system components via lead **190**. Different challenges exist for communicating electrical signals through a patient's body. For example, safety standards can limit the amount of current that can safely flow through a patient's body (particularly DC current). Additionally, the patient's body can act as an undesired signal path from component to component (e.g., via contact with the housing or “can” of each component). Various systems and methods can be employed to improve the communication ability between system components.

(49) FIG. **5A** is a high level electrical schematic showing communication between the implantable battery and/or communication module and the signal processor. In the illustrated embodiment, the implantable battery and/or communication module includes circuitry in communication with circuitry in the signal processor. Communication between the circuitry in the implantable battery and/or communication module and the signal processor can be facilitated by a lead (**190**), represented by the lead transfer function. The lead transfer function can include, for example, parasitic resistances and capacitances between the leads connecting the implantable battery and/or communication module and the signal processor and the patient's body and/or between two or more conductors that make up the lead (e.g., **191**). Signals communicated from the circuitry of the implantable battery and/or communication module to the circuitry in the signal processor can include electrical power provided to operate and/or stimulate system components (e.g., the middle ear sensor, signal processor, electrical and/or acoustic stimulator, and/or cochlear electrode) and/or data (e.g., processing data regarding the transfer function of the signal processor).

(50) As discussed elsewhere herein, the body of the patient provides an electrical path between system components, such as the “can” of the implantable battery and/or communication module and the “can” of the signal processor. This path is represented in FIG. **5A** by the flow path through R.sub.Body. Thus, the patient's body can provide undesirable signal paths which can negatively impact communication between components. To address this, in some embodiments, operating circuitry in each component can be substantially isolated from the component “can” and thus the patient's body. For example, as shown, resistance R.sub.Can is positioned between the circuitry and

the “can” of both the implantable battery and/or communication module and the signal processor.

(51) While being shown as R.sub.Can in each of the implantable battery and/or communication module and the signal processor, it will be appreciated that the actual value of the resistance between the circuitry and respective “can” of different elements is not necessarily equal.

Additionally, R.sub.Can need not include purely a resistance, but can include other components, such as one or more capacitors, inductors, and the like. That is, R.sub.Can can represent an insulating circuit including any variety of components that act to increase the impedance between circuitry within a component and the “can” of the component. Thus, R.sub.Can can represent an impedance between the operating circuitry of a component and the respective “can” and the patient's tissue. Isolating the circuitry from the “can” and the patient's body acts to similarly isolate the circuitry from the “can” of other components, allowing each component to operate with reference to a substantially isolated component ground. This can eliminate undesired communication and interference between system components and/or between system components and the patient's body.

(52) For example, as described elsewhere herein, in some examples, an electrical stimulator can provide an electrical stimulus to one or more contact electrodes on a cochlear electrode implanted in a patient's cochlear tissue. FIG. 5B illustrates an exemplary schematic diagram illustrating a cochlear electrode having a plurality of contact electrodes and fixedly or detachably connected to an electrical stimulator. As shown, the cochlear electrode **500** has four contact electrodes **502**, **504**, **506**, and **508**, though it will be appreciated that any number of contact electrodes is possible. As described elsewhere herein, the electrical stimulator can provide electrical signals to one or more such contact electrodes in response to an output from the signal processor according to the transfer function thereof and a received input signal.

(53) Because each contact electrode **502-508** is in contact with the patient's cochlear tissue, each is separated from the “can” of the electrical stimulator (as well as the “cans” of other system components) via the impedance of the patient's tissue, shown as R.sub.Body. Thus, if the circuitry within various system components did not have sufficiently high impedance (e.g., R.sub.Can) to the component “can”, electrical signals may stimulate undesired regions of the patient's cochlear tissue. For instance, stimulation intended for a particular contact electrode (e.g., **502**) may lead to undesired stimulation of other contact electrodes (e.g., **504**, **506**, **508**), reducing the overall efficacy of the system. Minimizing the conductive paths between system components (e.g., to the contact electrodes of a cochlear electrode) due to the patient's body, such as by incorporating impedances between component circuitry and the corresponding “can” via R.sub.Can, can therefore improve the ability to apply an electrical stimulus to only a desired portion of the patient's body.

(54) It will be appreciated that the term R.sub.Body is used herein to generally represent the resistance and/or impedance of the patient's tissue between various components and does not refer to a specific value. Moreover, each depiction of R.sub.Body in the figures does not necessarily represent the same value of resistance and/or impedance as the others.

(55) FIG. 6A shows a high level schematic diagram illustrating an exemplary communication configuration between an implantable battery and/or communication module, a signal processor, and a stimulator. In the example of FIG. 6A, the implantable battery and/or communication module **610** is in two-way communication with the signal processor **620**. For instance, the implantable battery and/or communication module **610** can communicate power and/or data signals **650** to the signal processor **620**. In some examples, the power and data signals **650** can be included in a single signal generated in the implantable battery and/or communication module **610** and transmitted to the signal processor **620**. Such signals can include, for example, a digital signal transmitted with a particular clock rate, which in some embodiments, can be adjustable, for example, via the implantable battery and/or communication module **610**.

(56) In some embodiments, the signal processor **620** can communicate information to the implantable battery and/or communication module **610** (e.g., **651**), for example, feedback

information and/or requests for more power, etc. The implantable battery and/or communication module **610** can, in response, adjust its output to the signal processor **620** (e.g., an amplitude, duty cycle, clock rate, etc.) in order to accommodate for the received feedback (e.g., to provide more power, etc.). Thus, in some such examples, the implantable battery and/or communication module **610** can communicate power and data (e.g., **650**) to the signal processor **620**, and the signal processor **620** can communicate various data back to the implantable battery and/or communication module **610** (e.g., **651**).

(57) In some embodiments, similar communication can be implemented between the signal processor **620** and the stimulator **630**, wherein the signal processor **620** provides power and data to the stimulator **630** (e.g., **660**) and receives data in return from the stimulator **630** (e.g., **661**). For example, the signal processor **620** can be configured to output signals (e.g., power and/or data) to the stimulator **630** (e.g., based on received inputs from a middle ear sensor or other device) via a similar communication protocol as implemented between the implantable battery and/or communication module **610** and the signal processor **620**. Similarly, in some embodiments, the stimulator can be configured to provide feedback signals to the signal processor, for example, representative of an executed stimulation process. Additionally or alternatively, the stimulator may provide diagnostic information, such as electrode impedance and neural response telemetry or other biomarker signals.

(58) FIG. **6B** is a schematic diagram illustrating exemplary electrical communication between an implantable battery and/or communication module and a signal processor in a cochlear implant system according to some embodiments. In the illustrated embodiment, the implantable battery and/or communication module **610** includes a signal generator **612** configured to output a signal through a lead (e.g., **190**) to the signal processor **620**. As described with respect to FIG. **6A**, in some examples, the signal generator **612** is configured to generate both data and power signals (e.g., **650**) for communication to the signal processor **620**. In some embodiments, the signal generator **612** generates a digital signal for communication to the signal processor **620**. The digital signal from the signal generator **612** can be communicated to the signal processor **620** at a particular clock rate. In some examples, the signals are generated at approximately 30 kHz. In various examples, data and power frequencies can range from approximately 100 Hz to approximately 10 MHz, and in some examples, may be adjustable, for example, by a user.

(59) In the illustrated embodiment, the implantable battery and/or communication module **610** includes a controller in communication with the signal generator **612**. In some examples, the controller is capable of adjusting communication parameters such as the clock rate of the signal generator **612**. In an exemplary embodiment, the controller and/or the signal generator **612** can communicate with, for example, a patient's external programmer (e.g., as shown in FIG. **1**). The controller and/or signal generator **612** can be configured to communicate data to the signal processor **620** (e.g., **651**), such as updated firmware, signal processor **620** transfer functions, or the like.

(60) As shown, the signal generator **612** outputs the generated signal to an amplifier **690** and an inverting amplifier **692**. In some examples, both amplifiers are unity gain amplifiers. In some examples comprising digital signals, the inverting amplifier **692** can comprise a digital NOT gate. The output from the amplifier **690** and the inverting amplifier **692** are generally opposite one another and are directed to the signal processor **620**. In some embodiments, the opposite nature of the signals output to the signal processor **620** from amplifiers **690** and **692** results in a charge-neutral communication between the implantable battery and/or communication module **610** and the signal processor **620**, such that no net charge flows through the wearer.

(61) In the illustrated example of FIG. **6B**, the receiving circuitry in the signal processor **620** comprises a rectifier circuit **622** that receives signals (e.g., **650**) from the amplifier **690** and the inverting amplifier **692**. Since the output of one of the amplifiers **690** and **692** will be high, the rectifier circuit **622** can be configured to receive the opposite signals from the amplifiers **690** and

692 and generate therefrom a substantially DC power output **623**. In various embodiments, the DC power **623** can be used to power a variety of components, such as the signal processor **620** itself, the middle ear sensor, the electrical and/or acoustic stimulator, or the like. The rectifier circuit **622** can include any known appropriate circuitry components for rectifying one or more input signals, such as a diode rectification circuit or a transistor circuit, for example.

(62) As described elsewhere herein, the implantable battery and/or communication module **610** can communicate data to the signal processor **620**. In some embodiments, the controller and/or the signal generator **612** is configured to encode the data for transmission via the output amplifiers **690** and **692**. The signal processor **620** can include a signal extraction module **624** configured to extract the data signal **625** from the signal(s) (e.g., **650**) communicated to the signal processor **620** to produce a signal for use by the signal processor **620**. In some examples, the signal extraction module **624** is capable of decoding the signal that was encoded by the implantable battery and/or communication module **610**. Additionally or alternatively, the signal extraction module **624** can extract a signal **625** resulting from the lead transfer function. In various examples, the extracted signal **625** can include, for example, an updated transfer function for the signal processor **620**, a desired stimulation command, or other signals that affect operation of the signal processor **620**.

(63) In the illustrated example, the signal processor **620** includes a controller **626** that is capable of monitoring the DC power **623** and the signal **625** received from the implantable battery and/or communication module **610**. The controller **626** can be configured to analyze the received DC power **623** and the signal **625** and determine whether or not the power and/or signal is sufficient. For example, the controller **626** may determine that the signal processor **620** is receiving insufficient DC power for stimulating a cochlear electrode according to the signal processor **620** transfer function, or that data from the implantable battery and/or communication module **610** is not communicated at a desired rate. Thus, in some examples, the controller **626** of the signal processor **620** can communicate with the controller **614** of the implantable battery and/or communication module **610** and provide feedback regarding the received communication. Based on the received feedback from the controller **626** of the signal processor **620**, the controller **614** of the implantable battery and/or communication module **610** can adjust various properties of the signal output by the implantable battery and/or communication module **610**. For example, the controller of the implantable battery and/or communication module **610** can adjust the clock rate of the communication from the signal generator **612** to the signal processor **620**.

(64) In some systems, the transmission efficiency between the implantable battery and/or communication module **610** and the signal processor **620** is dependent on the clock rate of transmission. Accordingly, in some examples, the implantable battery and/or communication module **610** begins by transmitting at an optimized clock rate until a change in clock rate is requested via the signal processor **620**, for example, to enhance data transmission (e.g., rate, resolution, etc.). In other instances, if more power is required (e.g., the controller of the signal processor **620** determines the DC power is insufficient), the clock rate can be adjusted to improve transmission efficiency, and thus the magnitude of the signal received at the signal processor **620**. It will be appreciated that in addition or alternatively to adjusting a clock rate, adjusting an amount of power transmitted to the signal processor **620** can include adjusting the magnitude of the signal output from the signal generator **612**. In some embodiments, for example, with respect to FIGS. **6A-B**, power and data can be communicated, for example, from implantable battery and/or communication module **610** to the signal processor **620** at a rate of approximately 30 kHz, and can be adjusted from there as necessary and/or as requested, for example, by the signal processor **620**.

(65) FIG. **7A** is an alternative high-level schematic diagram illustrating an exemplary communication configuration between an implantable battery and/or communication module, a signal processor, and a stimulator. In the example of FIG. **7A**, the implantable battery and/or communication module **710** provides signals (e.g., **750**) to the signal processor **720** via a first communication link and is further in two-way communication for providing additional signals

(e.g., 751) with the signal processor 720. In the example of FIG. 7A, the implantable battery and/or communication module 710 can provide power signals (e.g., 750) to the signal processor 720 via a communication link and otherwise be in two-way data communication (751) with the signal processor 720 via a second communication link. In some such examples, the power (750) and data (751) signals can each include digital signals. However, in some embodiments, the power and data signals are transmitted at different clock rates. In some examples, the clock rate of the data signals is at least one order of magnitude greater than the clock rate of the power signals. For example, in an exemplary embodiment, the power signal is communicated at a clock rate of approximately 30 kHz, while the data communication occurs at a clock rate of approximately 1 MHz. Similarly to the embodiment described in FIG. 6A, in some examples, the clock rate can be adjustable, for example, via the implantable battery and/or communication module 710.

(66) As described with respect to FIG. 6A, in some embodiments, the signal processor 720 can communicate information to the implantable battery and/or communication module 710, for example, feedback information and/or requests for more power, etc. (e.g., data signals 751). The implantable battery and/or communication module 710 can, in response, adjust the power and/or data output to the signal processor 720 (e.g., an amplitude, duty cycle, clock rate, etc.) in order to accommodate for the received feedback (e.g., to provide more power, etc.).

(67) In some embodiments, similar communication can be implemented between the signal processor 720 and the stimulator 730, wherein the signal processor 720 provides power and data to the stimulator 730 and receives data in return from the stimulator 730. For example, the signal processor 720 can be configured to output signals power signals (e.g., 760) and data signals (e.g., 761) to the stimulator 730 (e.g., based on received inputs from a middle ear sensor or other device). Such communication can be implemented via a similar communication protocol as implemented between the implantable battery and/or communication module 710 and the signal processor 720. In some examples, the power signals provided to the stimulator 730 (e.g., 760) are the same signals (e.g., 750) received by the signal processor 720 from the implantable battery and/or communication module 710. Additionally, in some embodiments, the stimulator 730 can be configured to provide feedback signals to the signal processor 720 (e.g., 761), for example, representative of an executed stimulation process.

(68) FIG. 7B is an alternative schematic diagram illustrating exemplary electrical communication between an implantable battery and/or communication module 710b and a signal processor 720b in a cochlear implant system similar to that shown in FIG. 7A. In the illustrated embodiment of FIG. 7B, the implantable battery and/or communication module 710b includes a power signal generator 711 and a separate signal generator 712. The power signal generator 711 and signal generator 712 are each configured to output a signal through a lead (e.g., 190) to the signal processor 720b. In some embodiments, the power signal generator 711 and the signal generator 712 each generates digital signal for communication to the signal processor 720b. In some such embodiments, the digital signal (e.g., 750) from the power signal generator 711 can be communicated to the signal processor 720b at a power clock rate, while the digital signal (e.g., 751b) from the signal generator 712 can be communicated to the signal processor 720b at a data clock rate that is different from the power clock rate. For instance, in some configurations, power and data can be communicated most effectively and/or efficiently at different clock rates. In an exemplary embodiment, the power clock rate is approximately 30 kHz while the data clock rate is approximately 1 MHz. Utilizing different and separately communicated power and data signals having different clock rates can increase the transfer efficiency of power and/or data from the implantable battery and/or communication module 710b to the signal processor 720b.

(69) In the illustrated embodiment, the implantable battery and/or communication module 710b includes a controller 714 in communication with the power signal generator 711 and the signal generator 712. In some examples, the controller 714 is capable of adjusting communication parameters such as the clock rate or content of the signal generator 712 and/or the power signal

generator **711**. In an exemplary embodiment, the controller **714** and/or the signal generator **712** or power signal generator **711** can communicate with, for example, a patient's external programmer (e.g., as shown in FIG. **1**). The controller **714** and/or signal generator **712** can be configured to communicate data to the signal processor **720b**, such as updated firmware, signal processor **720b** transfer functions, or the like. Additionally or alternatively, the controller **714** can be configured to transmit signals such as audio or other signals streamed or otherwise received from one or more external devices as described elsewhere herein.

(70) As shown, and similar to the example shown in FIG. **6B**, the power signal generator **711** outputs the generated signal to an amplifier **790** and an inverting amplifier **792**. In some examples, both amplifiers are unity gain amplifiers. In some examples comprising digital signals, the inverting amplifier **792** can comprise a digital NOT gate. The output from the amplifier **790** and the inverting amplifier **792** are generally opposite one another and are directed to the signal processor **720b**. In the illustrated example, the receiving circuitry in the signal processor **720b** comprises a rectifier circuit **722** that receives signals from the amplifier **790** and the inverting amplifier **792**. Since the output of one of the amplifiers **790** and **792** will be high, the rectifier circuit **722** can be configured to receive the opposite signals from the amplifiers **790** and **792** and generate therefrom a substantially DC power output **723**.

(71) In various embodiments, the DC power **723** can be used to power a variety of components, such as the signal processor **720b** itself, the middle ear sensor, the electrical and/or acoustic stimulator **730**, or the like. The rectifier circuit **722** can include any known appropriate circuitry components for rectifying one or more input signals, such as a diode rectification circuit or a transistor circuit, for example. In some embodiments, signals from the power signal generator **711** are generated at a clock rate that is optimal for transmitting power through the lead (e.g., approximately 30 kHz). In the illustrated example of FIG. **7B**, the rectifier circuit **722** can be arranged in parallel with power lines that are configured to communicate power signals to other components within the system, such as the stimulator **730**, for example. For instance, in some embodiments, the same power signal (e.g., **750**) generated from the power signal generator **711** and output via amplifiers **790** and **792** can be similarly applied to the stimulator **730**. In some such examples, the stimulator **730** includes a rectifier circuit **722** similar to the signal processor **720b** for extracting DC power from the power signal and the inverted power signal provided by amplifiers **790** and **792**, respectively. In alternative embodiments, the signal processor **720b** can similarly provide signals from a separate power signal generator **711** to provide power signals (e.g., at approximately 30 kHz) to the stimulator **730** similar to how power is provided from the implantable battery and/or communication module **710b** to the signal processor **720b** in FIG. **7B**.

(72) In the example of FIG. **7B**, the signal generator **712** outputs a data signal (e.g., **751b**) to an amplifier **794** and an inverting amplifier **796**. In some examples, both amplifiers are unity gain amplifiers. In some examples comprising digital signals, the inverting amplifier **796** can comprise a digital NOT gate. The output from the amplifier **794** and the inverting amplifier **796** are generally opposite one another and are directed to the signal processor **720b**.

(73) As described elsewhere herein, in some embodiments, the controller **714** and/or the signal generator **712** is configured to encode data for transmission via the output amplifiers **794** and **796**. The signal processor **720b** can include a signal extraction module **724** configured to extract the data from the signal(s) **725** communicated to the signal processor **720b** to produce a signal **725** for use by the signal processor **720b**. In some examples, the signal extraction module **724** is capable of decoding the signal that was encoded by the implantable battery and/or communication module **710b**. Additionally or alternatively, the signal extraction module **724** can extract a resulting signal **725** resulting from the lead transfer function. In various examples, the extracted signal can include, for example, an updated transfer function for the signal processor **720b**, a desired stimulation command, or other signals that affect operation of the signal processor **720b**.

(74) In the example of FIG. **7B**, the signal extraction module **724** includes a pair of tri-state buffers

786 and **788** in communication with signals output from the signal generator **712**. The tri-state buffers **786** and **788** are shown as having “enable” (ENB) signals provided by controller **726** in order to control operation of the tri-state buffers **786** and **788** for extracting the signal from the signal generator **712**. Signals from the signal generator **712** and buffered by tri-state buffers **786** and **788** are received by amplifier **784**, which can be configured to produce a signal **725** representative of the signal generated by the signal generator **712**.

(75) In some examples, communication of signals generated at the signal generator **712** can be communicated to the signal processor **720b** at a clock rate that is different from the clock rate of the signals generated by the power signal generator **711**. For instance, in some embodiments, power signals from the power signal generator **711** are transmitted at approximately 30 kHz, which can be an efficient frequency for transmitting power. However, in some examples, the signals from the signal generator **712** are transmitted at a higher frequency than the signal from the power signal generator **711**, for example, at approximately 1 MHz. Such high frequency data transmission can be useful for faster data transfer than would be available at lower frequencies (e.g., the frequencies for transmitting the signal from the power signal generator **711**). Thus, in some embodiments, power and data can be communicated from the implantable battery and/or communication module **710b** to the signal processor **720b** via different communication channels at different frequencies.

(76) Similar to the embodiment shown in FIG. 6B, in the illustrated example of FIG. 7B, the signal processor **720b** includes a controller **726** that is in communication with the implantable battery and/or communication module **710b**. In some such embodiments, the controller **726** in the signal processor **720b** is capable of monitoring the DC power **723** and/or the signal **725** received from the implantable battery and/or communication module **710b**. The controller **726** can be configured to analyze the received DC power **723** and the signal **725** and determine whether or not the power and/or signal is sufficient. For example, the controller **726** may determine that the signal processor **720b** is receiving insufficient DC power for stimulating a cochlear electrode according to the signal processor **720b** transfer function, or that data from the implantable battery and/or communication module **710b** is not communicated at a desired rate. Thus, in some examples, the controller **726** of the signal processor **720b** can communicate with the controller **714** of the implantable battery and/or communication module **710b** and provide feedback regarding the received communication. Based on the received feedback from the controller **726** of the signal processor **720b**, the controller **714** of the implantable battery and/or communication module **710b** can adjust various properties of the signals output by the power signal generator **711** and/or the signal generator **712**.

(77) In the illustrated example of FIG. 7B, bidirectional communication signals **751b** between the implantable battery and/or communication module **710b** and signal processor **720b** comprises signals from the amplifiers **794** and **796** in one direction, and communication from controller **726** to controller **714** in the other direction. It will be appreciated that a variety of communication protocols and techniques can be used in establishing bidirectional communication signals **751b** between the implantable battery and/or communication module **710b** and signal processor **720b**.

(78) For example, in some embodiments, the implantable battery and/or communication module **710b** need not include amplifiers **794** and **796**, and instead transmits a signal and not its inverse to the signal processor **720b**. In other examples, the signal processor includes amplifiers similar to **794** and **796**, and outputs a signal and its inverse back to the implantable battery and/or communication module **710b**. Additionally or alternatively, in some embodiments, the signal generator **712** can be integral with the controller **714** and/or the signal extraction module **724** can be integral with controller **726**, wherein controllers **714** and **726** can be in bidirectional communication via signal generator **712** and/or the signal extraction module **724**. In general, the implantable battery and/or communication module **710b** and the signal processor **720b** can be in bidirectional communication for communicating data signals separate from the power signals provided by power signal generator **711**.

(79) As described, separate communication channels for power (e.g., **750**) and data (e.g., **751b**) can

be used for providing both power and data from the implantable battery and/or communication module **710b** and the signal processor **720b**. This can allow for separate data and power clocking rates in order to improve the power transmission efficiency as well as the data transmission efficiency and/or rate. Moreover, in some examples, if the bidirectional communication (e.g., **751b**) between the implantable battery and/or communication module **710b** and the signal processor **720b** fails (e.g., due to component failure, connection failure, etc.), data for communication from the implantable battery and/or communication module **710b** can be encoded in the power signals (e.g., **750**) from the power signal generator **711** and transmitted to the signal processor **720b**. Thus, similar to the embodiment described with respect to FIG. 6B, both power and data can be transmitted via the same signal.

(80) In some examples, the signal extraction module **724** can be configured to receive data received from the power signal generator **711**, for example, via an actuatable switch that can be actuated upon detected failure of communication **751b**. In other examples, the signal extraction module **724** and/or the controller **726** can generally monitor data from the power signal generator **711** and identify when signals received from the power signal generator **711** include data signals encoded into the received power signal in order to determine when to consider the power signals to include data.

(81) Accordingly, in some embodiments, the configuration of FIG. 7B can be implemented to establish efficient, bidirectional communication between the implantable battery and/or communication module **710b** and the signal processor **720b**. Failure in bidirectional communication **751b** can be identified manually and/or automatically. Upon detection of failure in the bidirectional communication **751b**, the controller **714** can encode data into the power signal output from the power signal generator **711**, and power and data can be combined into a single signal such as described with respect to FIG. 6B.

(82) FIG. 7C is another alternative schematic diagram illustrating exemplary electrical communication between an implantable battery and/or communication module **710c** and a signal processor **720c** in a cochlear implant system similar to that shown in FIG. 7A. Similar to the embodiment of FIG. 7B, in the illustrated embodiment of FIG. 7C, the implantable battery and/or communication module **710c** includes a power signal generator **711** configured to output a signal through a lead (e.g., **190**) to the signal processor **720c**. In some embodiments, the power signal generator **711** generates a digital signal (e.g., **750**) for communication to the signal processor **720c**, for example, at a power clock rate. The power signal generator **711** and corresponding amplifiers **790**, **792**, as well as rectifier circuit **722**, can operate similar to described with respect to FIG. 7B in order to extract DC power **723** and, in some examples, output power signals to further system components, such as stimulator **730**.

(83) In the illustrated embodiment, the implantable battery and/or communication module **710c** includes a signal generator **713**, which can be capable of providing data signals to the signal processor. In some embodiments, the signal generator **713** generates a digital signal for communication to the signal processor **720c**. In some such embodiments, the digital signal (e.g., **751c**) from the signal generator **713** can be communicated to the signal processor **720b** at a data clock rate that is different from the power clock rate. For instance, as described elsewhere herein, in some configurations, power and data can be communicated most effectively and/or efficiently at different clock rates. In an exemplary embodiment, the power clock rate is approximately 30 kHz while the data clock rate is approximately 1 MHz. Utilizing different and separately communicated power and data signals having different clock rates can increase the transfer efficiency of power and/or data from the implantable battery and/or communication module **710c** to the signal processor **720c**.

(84) The embodiment of FIG. 7C includes a controller **715** in communication with the power signal generator **711** and the signal generator **713**. In some examples, the controller **715** is capable of adjusting communication parameters such as the clock rate or content of the signal generator **713**

and/or the power signal generator **711**. In an exemplary embodiment, the controller **715** and/or the signal generator **713** or power signal generator **711** can communicate with, for example, a patient's external programmer (e.g., as shown in FIG. **1**). The controller **715** and/or signal generator **713** can be configured to communicate data to the signal processor **720c**, such as updated firmware, signal processor **720c** transfer functions, or the like.

(85) Similar to the example in FIG. **7B**, in the example of FIG. **7C**, the signal generator **713** outputs a data signal (e.g., **751**) to an amplifier **795** and an inverting amplifier **797**. In some examples, both amplifiers are unity gain amplifiers. In some examples, amplifiers **795**, **797** comprise tri-state buffers. In some examples comprising digital signals, the inverting amplifier **797** can comprise a digital NOT gate. The output from the amplifier **795** and the inverting amplifier **797** are generally opposite one another and are directed to the signal processor **720c**.

(86) As described elsewhere herein, in some embodiments, the controller **715** and/or the signal generator **713** is configured to encode data for transmission via the amplifiers **795** and **797**. The signal processor **720c** can include a signal extraction module **734** configured to extract the data from the signal(s) communicated to the signal processor **720c** to produce a signal for use by the signal processor **720c**. In some examples, the signal extraction module **734** is capable of decoding the signal that was encoded by the implantable battery and/or communication module **710c**.

Additionally or alternatively, the signal extraction module **734** can extract a signal resulting from the lead transfer function. In various examples, the extracted signal can include, for example, an updated transfer function for the signal processor **720c**, a desired stimulation command, or other signals that affect operation of the signal processor **720c**.

(87) In the example of FIG. **7C**, similar to signal extraction module **724** in FIG. **7B**, the signal extraction module **734** includes a pair of tri-state buffers **787** and **789** in communication with signals output from the signal generator **713**. The tri-state buffers **787** and **789** are shown as having “enable” (ENB) signals provided by controller **727** in order to control operation of the tri-state buffers **787** and **789** for extracting the signal from the signal generator **713**. Signals from the signal generator **713** and buffered by tri-state buffers **787** and **789** are received by amplifier **785**, which can be configured to produce a signal representative of the signal generated by the signal generator **713**.

(88) As described elsewhere herein, in some examples, communication of signals generated at the signal generator **713** can be communicated to the signal processor **720c** at a clock rate that is different from the clock rate of the signals generated by the power signal generator **711**. For instance, in some embodiments, power signals from the power signal generator **711** are transmitted at approximately 30 kHz, which can be an efficient frequency for transmitting power. However, in some examples, the signals from the signal generator **713** are transmitted at a higher frequency than the signal from the power signal generator **711**, for example, at approximately 1 MHz. Such high frequency data transmission can be useful for faster data transfer than would be available at lower frequencies (e.g., the frequencies for transmitting the signal from the power signal generator **711**). Thus, in some embodiments, power and data can be communicated from the implantable battery and/or communication module **710c** to the signal processor **720c** via different communication channels at different frequencies.

(89) In the illustrated example of FIG. **7C**, the signal processor **720c** includes a signal generator **717** and controller **727** that is in communication with the signal generator **717**. Similar to the operation of signal generator **713** and amplifiers **795** and **799**, the signal generator can be configured to produce output signals to buffers **787** and **789**, which can be configured to output signals to the implantable battery and/or communication module **710c**.

(90) In some embodiments, the controller **727** in the signal processor **720c** is capable of monitoring the DC power **723** and/or the signal received from the implantable battery and/or communication module **710c**. The controller **726** can be configured to analyze the received DC power **723** and the signal and determine whether or not the power and/or signal is sufficient. For example, the

controller **727** may determine that the signal processor **720c** is receiving insufficient DC power for stimulating a cochlear electrode according to the signal processor **720c** transfer function, or that data from the implantable battery and/or communication module **710c** is not communicated at a desired rate. Thus, in some examples, the controller **727** of the signal processor **720c** cause the signal generator **717** to generate communication signals to send to implantable battery and/or communication module **710c**. Such signals can be used to provide feedback regarding signals received by the signal processor **720c**, such as the DC power **723**.

(91) In the example of FIG. 7C, amplifiers **795** and **797** are shown as including tri-state amplifiers (e.g., tri-state buffers) controllable by the controller **727**. Similar to the configuration in the signal processor **720c**, the implantable battery and/or communication module **710c** includes a signal extraction module **735** configured to extract data from the signal(s) communicated to the implantable battery and/or communication module **710c** from signal generator **717** of the signal processor **720c**. The signal extraction module **735** includes amplifiers **795** and **797** (e.g., tri-state buffers) in communication with signals output from the signal generator **717**. Signals from the signal generator **717** and received at amplifiers **795** and **797** are received by amplifier **799**, which can be configured to produce a signal representative of the signal generated by the signal generator **717** to controller **715** of the implantable battery and/or communication module **710**. Thus, in some embodiments, the controller **727** of the signal processor **720c** is configured to communicate data back to the implantable battery and/or communication module **710a** via buffers **787** and **789**.

(92) As described with respect to other embodiments, based on the received feedback from the controller **727** of the signal processor **720c**, the controller **715** of the implantable battery and/or communication module **710c** can adjust various properties of the signals output by the power signal generator **711** and/or the signal generator **713**.

(93) Thus, in the illustrated example of FIG. 7C, bidirectional communication signal **751** between the implantable battery and/or communication module **710c** and signal processor **720c** includes communication between different signal extraction modules **735** and **734**. As shown, both the implantable battery and/or communication module **710c** and the signal processor **720c** include a controller (**715**, **727**) that communicates with a signal generator (**713**, **717**) for producing output signals. The signal generator (**713**, **717**) outputs signals via tri-state amplifiers, including one inverting amplifier (**797**, **789**) for communication across bidirectional communication **751c** for receipt by the other signal extraction module (**734**, **735**).

(94) Thus, in some embodiments, bidirectional communication **751c** between the implantable battery and/or communication module **710c** and the signal processor **720c** can be enabled by each of the implantable battery and/or communication module and the signal processor receiving and transmitting data via approximately the same communication structure as the other. In some such examples, the implantable battery and/or communication module **710c** and the signal processor **720c** include signal extraction modules **735** and **734**, respectively, configured both to output signals from a signal generator (e.g., via signal generator **713** or signal generator **717**) and receive and extract signals (e.g., via amplifier **785** and amplifier **799**).

(95) In the example of FIG. 7C, amplifiers **795** and **797** comprise tri-state amplifiers that selectively (e.g., via “enable” control from controller **715**) output the signal from signal generator **713**, and amplifier **797** is shown as an inverting amplifier. As described, in some examples, amplifiers **795** and **797** comprise tri-state buffers. Similarly, of tri-state buffers **787** and **789** that selectively (e.g., via “enable” control from controller **727**) output the signal from signal generator **717**, buffer **789** is shown as an inverting amplifier. As described elsewhere herein, communicating a signal and its inverse (e.g., via **795** and **797**) allows communication with no net charge flow between the implantable battery and/or communication module **710c** and the signal processor **720c**. Thus, bidirectional communication between the implantable battery and/or communication module **710c** and the signal processor **720c** can be performed without a net charge flow between the components.

(96) As described elsewhere herein, power from power generator **711** and data from signal generator **713** (and/or signal generator **717**) can be communicated at different clocking rates to optimize power and data transfer. In some examples, if data communication (e.g., via bidirectional communication **751c**) fails, the controller **715** can be configured to control power generator **711** to provide both power and data signals via amplifiers **790** and **792**, for example, as described with respect to FIG. **6B**.

(97) Accordingly, in some embodiments, the configuration of FIG. **7C** can be implemented to establish efficient, bidirectional communication between the implantable battery and/or communication module **710** and the signal processor **720**. Failure in bidirectional communication **751** can be identified manually and/or automatically. Upon detection of failure in the bidirectional communication **751**, the controller **715** can encode data into the power signal output from the power signal generator **711**, and power and data can be combined into a single signal such as described with respect to FIG. **6B**.

(98) As discussed elsewhere herein, different safety standards can exist regarding electrical communication within the patient's body. For example, safety standards can limit the amount of current that can safely flow through a patient's body (particularly DC current). As shown in FIGS. **6B**, **7B**, and **7C**, each of the illustrated communication paths between the implantable battery and/or communication module and the signal processor are coupled to output capacitors. The capacitors positioned at the inputs and outputs of the implantable battery and/or communication module and the signal processor can substantially block DC current from flowing therebetween while permitting communication of AC signals.

(99) As described elsewhere herein, in some embodiments, the data communicated between the implantable battery and/or communication module and the signal processor (e.g., from the signal generator) is encoded. In some such examples, the encoding can be performed according to a particular data encoding method, such as an 8b/10b encoding scheme, to achieve DC balance in the communicated signal. For example, in some embodiments, data is encoded such that the numbers of high and low bits communicated between components at each clock signal meet certain criteria to prevent a charge of a single polarity from building up on any of the capacitors. Such encoding can minimize the total charge that flows between the implantable battery and/or communication module and the signal processor during communication.

(100) While described and illustrated as representing communication between the implantable battery and/or communication module and the signal processor, it will be appreciated that communication configurations such as shown in FIGS. **5A**, **5B**, **6A**, **6B**, **7A**, **7B**, and **7C** can be implemented between any pair of devices generally in communication with one another. For example, isolating circuitry (e.g., R.sub.Can) can be included in any of the system components (e.g., middle ear sensor, acoustic stimulator, electrical stimulator, etc.) to effectively isolate the ground signals from each component from its respective can. Similarly, the exemplary capacitive AC coupling with DC blocking capacitors and DC balancing encoding as described elsewhere herein can be incorporated as the communication interface between any two communicating components.

(101) As described, data can be communicated from the implantable battery and/or communication module to the signal processor for a variety of reasons. In some examples, data is that communicated to the implantable battery and/or communication module from an external device, such as a programmer as shown in FIG. **1**. In an exemplary process, a programmer, such as a clinician's computer, can be used to communicate with a patient's fully implanted system via a communication configuration such as shown in FIG. **6B**, **7B**, or **7C**. For example, a programmer can communicate wirelessly (e.g., via Bluetooth or other appropriate communication technique) with the patient's implantable battery and/or communication module. Signals from the programmer can be sent from the implantable battery and/or communication module to the signal processor via the communication configurations of FIG. **6B**, **7B**, or **7C**.

(102) During such processes, a clinician can communicate with the signal processor, and, in some cases, with other components via the signal processor. For example, the clinician can cause the signal processor to actuate an electrical and/or an acoustic stimulator in various ways, such as using various electrical stimulation parameters, combinations of active contact electrodes, various acoustic stimulation parameters, and various combinations thereof. Varying the stimulation parameters in real time can allow the clinician and patient to determine effectiveness of different stimulation techniques for the individual patient. Similarly, the clinician can communicate with the signal processor to update transfer function. For example, the clinician can repeatedly update the transfer function signal processor while testing the efficacy of each one on the individual patient. In some examples, combinations of stimulation parameters and signal processor transfer functions can be tested for customized system behavior for the individual patient.

(103) In some embodiments, various internal properties of the system may be tested. For instance, various impedance values, such as a sensor impedance or a stimulator impedance can be tested such as described in U.S. Patent Publication No. 2015/0256945, entitled TRANSDUCER IMPEDANCE MEASUREMENT FOR HEARING AID, which is assigned to the assignee of the instant application, the relevant portions of which are incorporated by reference herein.

(104) Additionally or alternatively, various characteristics of individual leads can be analyzed. FIG. 7D is high-level schematic diagram illustrating exemplary electrical communication between an implantable battery and/or communication module and a signal processor in a cochlear implant system similar to that shown in FIG. 7A. In the simplified example of FIG. 7D, conductors **701**, **702**, **703**, and **704** extend between implantable battery and/or communication module **710d** and signal processor **720d**. In some examples, such conductors are included in a lead (e.g., lead **190**) extending between the implantable battery and/or communication module **710d** and signal processor **720d**. In the example of FIG. 7D, implantable battery and/or communication module **710d** includes controller **705** and signal processor **720d** includes controller **706**. Other internal components of the implantable battery and/or communication module **710d** and signal processor **720d** are not shown, though various configurations are possible, such as shown in FIG. 6B, 7B, or 7C.

(105) In some embodiments, one or both of controllers **705**, **706** can be configured to apply a test signal to one or more of conductors **701**, **702**, **703**, **704** in order to test one or more properties of such conductors. In an exemplary test process, a controller (e.g., **705**) can drive a signal (e.g., a sine wave or other shaped wave) across a conductor (e.g., **701**) and measure the sent current and the voltage at which the current is sent. From this information, the controller can determine conductor impedance, including integrity of the conductor (e.g., whether or not the conductor is broken). Similarly, a controller can be configured to ground a second conductor (e.g., **702**) while driving the test signal across a test conductor (e.g., **701**) in order to measure one or more electrical parameters between the two conductors (e.g., capacitance, impedance, etc.).

(106) During exemplary operation, a controller can be configured to apply a test signal to a first conductor (e.g., **701**) and ground a second conductor (e.g., **702**). The controller can be configured to apply a test signal at a plurality of frequencies (e.g., perform a frequency sweep) and measure impedance vs. frequency between the first conductor and the second, grounded conductor. In various examples, a controller can be configured to perform such tests using any two conductors **701**, **702**, **703**, **704**, to test for baseline values (e.g., when the system is in a known working condition) or to test for expected values (e.g., to compare to an established baseline). In different embodiments, the controller in the implantable battery and/or communication module **710d** (controller **705**) and/or the controller in the signal processor **720d** (controller **706**) can perform the grounding of one or more conductors and/or apply the test signal to one or more conductors.

(107) In some embodiments, such test processes can be performed automatically, for example, according to a programmed schedule. Additionally or alternatively, such test processes can be initiated manually, for example, by a wearer or a clinician, via an external device such as via a

programmer (e.g., **100**) or charger (e.g., **102**). The results of such processes can be stored in an internal memory for later access and analysis, and/or can output to an external device for viewing. In some examples, results and/or a warning can be output to an external device automatically in the event that one or more results deviates sufficiently from a baseline value. In various examples, sufficient variation from the baseline for triggering an output can be based on a percent variation from the baseline (e.g., greater than 1% deviation from the baseline, greater than 5% deviation, greater than 10% deviation, etc.). Additionally or alternatively, sufficient variation can include varying a certain number of standard deviations from the baseline (e.g., greater than one standard deviation, two standard deviations, etc.). In various embodiments, the amount of variation that triggers outputting the results and/or a warning is adjustable. Additionally or alternatively, such an amount can vary between different measurements.

(108) In some embodiments, one or more actions may be performed in response to the results of such an analysis. For instance, in an exemplary embodiment described with respect to FIG. 7B, if a test reveals an unexpected impedance on one of the signal conductors (e.g., from amplifier **794** or inverting amplifier **796**), such as an open circuit, the controller **714** may be configured to change operation of the system. For instance, controller **714** can be configured to adjust the output from power generator **711** in order to provide both power and data signals from the power generator **711**, such as described with respect to the configuration in FIG. 6B. In some examples, the controller **714** can be configured to transmit a signal to an external device signaling such a change in operation and/or alerting a wearer and/or clinician that one or more conductors may be damaged or otherwise not operational.

(109) While shown in several embodiments (e.g., FIGS. 1, 4, 6A, 7A) as being separate components connected by a lead (e.g., lead **180**), in some examples, the processor (e.g., **120**) and the stimulator (e.g., **130**) can be integrated into a single component, for example, within a hermetically sealed housing. Some such embodiments are described in U.S. patent application Ser. No. 16/797,388, filed Feb. 21, 2020, and which is assigned to the assignee of the instant application and is incorporated herein by reference.

(110) As described elsewhere herein, while many examples show a middle ear sensor being in communication with an implanted signal processor, in various embodiments, one or more additional or alternative input sources can be included. For instance, in some embodiments, a microphone can be implanted under a user's skin and can be placed in communication with the signal processor (e.g., via a detachable connector such as **171**). The signal processor can receive input signals from the implanted microphone and provide signals to the stimulator based on the received input signal and the signal processor transfer function.

(111) Additionally or alternatively, one or more system components can be configured to receive broadcast signals for converting into stimulation signals. FIG. 8 is a schematic system diagram showing an implantable system configured to receive broadcast signals from a broadcast device. As shown in the example of FIG. 8, a broadcast source **850** broadcasts a signal via communication link **860**. The communication link **860** can include communication via a variety of communication protocols, such as Wi-Fi, Bluetooth, or other known data transmission protocols. Broadcast source **850** can include any of a variety of components, such as a media source (e.g., television, radio, etc.), communication device (e.g., telephone, smartphone, etc.), a telecoil or other broadcast system (e.g., at a live performance), or any other source of audio signals that can be transmitted to an implanted system or to an external device of an implanted system (e.g., a system programmer, etc.).

(112) An implantable system including a programmer **800**, an implantable battery and/or communication module **810**, a signal processor **820**, and a stimulator **830** can generally receive the data from the broadcast source **850** via communication link **860**. In various embodiments, any number of components in the implantable system can include a receiving device, such as a telecoil, configured to receive broadcast signals for eventual conversion into stimulation signals.

(113) For instance, in some embodiments, programmer **800** can include a telecoil relay configured

to receive broadcast telecoil signals from a broadcast source **850**. The programmer can be configured to subsequently communicate a signal representative of the received broadcast signal to the implantable battery and/or communication module **810** and/or the signal processor **820**, e.g., via a Bluetooth communication. If the communication is received from the programmer **800** via the implantable battery and/or communication module **810**, the implantable battery and/or communication module **810** can communicate the signal to the signal processor, for example, as described in any of FIG. 6A, 6B, 7A, or 7C.

(114) In some such embodiments, the signal processor **820** can be configured to receive such signals from the implantable battery and/or communication module **810** and output stimulation signals to the stimulator **830** based on the received signals and the signal processor transfer function. In other examples, the signal processor **820** can include a telecoil relay or other device capable of receiving broadcast signals from the broadcast source **850**. In some such embodiments, the signal processor **820** processes the received signals according to the signal processor transfer function and outputs stimulations signals to the stimulator **830**.

(115) In some embodiments, the signal processor **820** can be in communication with a plurality of input sources, such as, for example, a combination of an implanted microphone, a middle ear sensor, and a broadcast source **850** (e.g., via the implantable battery and/or communication module **810**). In some such examples, the signal processor can be programmed with a plurality of transfer functions, each according to respective input sources. In such embodiments, the signal processor can identify which one or more input sources are providing input signals and process each such input signal according to the transfer function associated with its corresponding input source.

(116) In some examples, a signal processor **820** receiving a plurality of input signals from a corresponding plurality of input sources effectively combines the signals when producing a stimulation signal to the stimulator **830**. That is, in some embodiments, input sources are combined to form the stimulation signal from the signal processor **820**. In some such examples, a user may be able to mix the various received input signals in any way desired. For example, a user may choose to blend a variety of different input streams, such as an input from a middle ear sensor or other implanted device, a signal received from an external device (e.g., a telecoil relay, a Bluetooth connection such as to a smartphone, etc.), and the like. In an exemplary configuration, a user may elect to equally blend two input sources such that the stimulation signal is based 50% on a first input source and 50% on a second input source.

(117) Additionally or alternatively, a user may elect to effectively “mute” one or more input sources so that the signal processor **820** outputs stimulations signals based on input signals received from unmuted sources. Similarly, a user may be able to select a single source from which to process received input signals. For example, in some embodiments, a user may select to have signals received from broadcast source **850** processed and converted into stimulation signals while having signals received from, for example, a middle ear sensor, disregarded.

(118) In some examples, direct communication with the signal processor can be used to test the efficacy of a given signal processor transfer function and associated stimulation (e.g., acoustic or electrical) parameters. For example, the programmer can be used to disable input signals from a middle ear sensor or other input source and provide a customized signal to the signal processor to simulate a signal from the input source. The signal processor processes the received signal according to its transfer function and actuates the electrical stimulator and/or the acoustic stimulator accordingly. The processor can be used to test a variety of customized “sounds” to determine the efficacy of the signal processor transfer function for the given patient for each “sound.”

(119) Various features and functions of implantable systems have been described herein. As described, in various embodiments, system operation(s) can be adjusted based on communication with the implanted system from components located outside of the body while the system remains implanted. In some embodiments, the system may include any number of external devices capable

of interfacing with the system in a variety of ways.

(120) FIG. 9 is a schematic diagram illustrating possible communication between a variety of system components according to some embodiments of a fully implantable system. In the illustrated embodiment, implanted components (outlined in broken line) of a system include an implantable battery and/or communication module **910**, a signal processor **920**, and a stimulator **930**. Such implanted components can operate according to various examples as described herein in order to effectively stimulate a user (e.g., via electrical and/or acoustic stimulation) in response to received input signals.

(121) The schematic illustration of FIG. 9 includes a plurality of external devices capable of wirelessly interfacing with one or more of the implanted components, for example, via communication link **925**. Such devices can include a programmer **900**, a charger **902**, a smartphone/tablet **904**, a smartwatch or other wearable technology **906**, and a fob **908**. In some examples, such components can communicate with one or more implantable components via one or more communication protocols via wireless communication link **925**, such as Bluetooth, Zigbee, or other appropriate protocols. In various embodiments, different external devices are capable of performing one or more functions associated with system operation. In some such embodiments, each external device is capable of performing the same functions as the others. In other examples, some external devices are capable of performing more functions than others.

(122) For example, a programmer **900** can be capable of interfacing wirelessly with one or more implantable components in order to control a variety of operating parameters of the implanted system. For example, in some embodiments, programmer **900** can be configured to adjust a signal processor transfer function or select an operating profile (e.g., associated with a particular signal processor transfer function according to a particular user, environment, etc.). In some examples, the programmer **900** can be used to establish user profiles, such as preferred signal processor transfer functions, as described elsewhere herein. The programmer **900** can additionally or alternatively be used to turn the system on or off, adjust the volume of the system, receive and stream input data to the system (e.g., the implantable battery and/or communication module **910**). In some embodiments, the programmer **900** includes a display for displaying various information to the user. For example, the display can be used to indicate a mode of operation (e.g., a loaded user profile), a remaining power level, or the like. In some such embodiments, the display can function as a user interface by which a user can adjust one or more parameters, such as volume, profile, input source, input mix, and the like.

(123) In some embodiments, a charger **902** can be used to charge one or more internal batteries or other power supplies within the system, such as in the implantable battery and/or communication module **910**. In some examples, the charger **902** can include the same functionality as the programmer **900**, including, for instance, a display and/or user interface. In some such embodiments, the programmer **900** and the charger **902** can be integrated into a single device.

(124) In some embodiments, various external devices such as a smartphone or tablet **904** can include an application (“app”) that can be used to interface with the implanted system. For example, in some embodiments, a user may communicate (e.g., via link **925**) with the system via the smartphone or tablet **904** in order to adjust certain operating factors of the system using a predefined app to provide an interface (e.g., a visual interface via a display integrated into the external device). The app can assist the user in adjusting various parameters, such as volume, operating profile, on/off, or the like. In some examples, the smartphone/tablet **904** can be used to stream input signals to the implanted system, such as media or communication playing on the smartphone/tablet **904**.

(125) In some systems, a smartwatch or other wearable technology **906** can interact with the system in a similar way as the smartphone/tablet **904**. For example, the smartwatch or other wearable technology **906** can include an app similar to that operable on the smartphone/tablet to control operation of various aspects of the implanted system, such as volume control, on/off control, etc.

(126) In some embodiments, the fob **908** can be used to perform basic function with respect to the implanted system. For instance, in some embodiments, a fob **908** can be used to load/implement a particular operating profile associated with the fob **908**. Additionally or alternatively, the fob **908** can function similar to the shut-off controller **104** of FIG. **1** and can be used to quickly disable and/or mute the system. As described elsewhere herein, in some examples, the same device used to disable and/or mute the system (e.g., fob **908**) can be used to enable and/or unmute the system.

(127) The schematic diagram of FIG. **9** further includes a broadcast source **950** configured to broadcast signals **960** that are receivable via one or more external devices and/or one or more implanted system components. Similar to the broadcast source **850** in FIG. **8**, broadcast source **950** can be configured to emit signals that can be turned into stimulation signals for application by stimulator **930**. Broadcast signals **960** can include, for example, telecoil signals, Bluetooth signals, or the like. In various embodiments, one or more external devices, such as a programmer **900**, charger **902**, smartphone/tablet **904**, smartwatch/wearable device **906**, and/or fob **908** can include a component (e.g., a telecoil relay) capable of receiving broadcast signal **960**. The external device(s) can be further configured to communicate a signal to one or more implanted components representative of the received broadcast signal **960** for applying stimulation to the patient based on the broadcast signal **960**.

(128) Additionally or alternatively, in some embodiments, one or more implanted system components, such as an implantable battery and/or communication module **910**, a signal processor **920**, and/or a stimulator **930** can be configured to receive broadcast signals **960**. Such component(s) can be used to generate stimulation signals for applying to a user via stimulator **930** according to the received broadcast signals **960**.

(129) As described, in some embodiments, various devices can communicate with components in an implanted system via wireless communication protocols such as Bluetooth. Various data and signals can be communicated wirelessly, including control signals and streaming audio. However, in some cases, such wireless communication should be made secure so that a system only communicates with those devices desired by the wearer. This can prevent unwanted signals from being broadcast to an implanted device and/or unauthorized access to one or more adjustable device settings.

(130) In some embodiments, one or more implanted system components comprises a near field communication component configured to facilitate communication between the system and an external device only when brought into very close proximity to the near field communication component. In some such examples, once near-field communication is established, the pairing for longer-range wireless communication (e.g., Bluetooth) can be established. For instance, in an exemplary embodiment, a charger and an implantable battery and/or communication module can each include near field communication components for establishing a secure, near field communication and subsequently pairing to each other for additional wireless communication.

(131) In embodiments wherein the external device includes, or is in communication with, a microphone, the external device can be configured to reprogram the signal processor based on information collected from the microphone representative of the acoustic environment. For example, the external device can be configured to identify background noise (e.g. low-end noise) and update the signal processor transfer function accordingly. In some such examples, the external device can be configured to reduce gain for low-end signals and/or emphasize other sounds or frequency ranges, such as speech or other sounds having a higher frequency. In some embodiments, a user can initiate the process of identifying background noise for adjusting the operation of the signal processor via the external device, for example, via a user interface (e.g., a smartphone or tablet touchscreen).

(132) In embodiments in which the external device includes or is in communication with a location sensor and/or a clock, the external device may reprogram the signal processor based on a detected location and/or time. For instance, in an example embodiment, when the external device is located

in a place known to be loud (e.g. a mall or sports stadium), the external device can be configured to detect the location and automatically reprogram the signal processor to reduce background noise (e.g., a particular frequency or range of frequencies) and/or reduce the overall gain associated with the transfer function. Similarly, in some examples, when located in a place in which a wearer may wish to particularly recognize speech (e.g., a movie theater) the external device can be configured to reprogram the signal processor to emphasize frequencies associated with speech.

(133) In some examples, the transfer function can be updated to reduce a contribution of identified background noise. In some embodiments, reducing a contribution of identified background noise comprises emphasizing signals having frequency content between approximately 200 Hz and 20 kHz. In some such examples, updating the transfer function to reduce a contribution of the identified background noise comprises emphasizing signals having frequency content between approximately 300 Hz and 8 kHz. Emphasizing signals in such frequency ranges can help emphasize human speech or other similar signals within a noisy environment.

(134) Additionally or alternatively, the external device can be configured to reprogram the signal processor based on a determined time of day. For example, at times when the wearer generally doesn't want to be bothered (e.g. at night), the external device can be configured to lower the volume of all or most sounds. In some examples, the wearer may additionally or alternatively temporarily reprogram the signal processor via the external device to adjust the transfer function of the signal processor (e.g., to reduce volume) for a predetermined amount of time (e.g. 15 minutes, 1 hour, or 1 day).

(135) In some examples, reprogramming the signal processor comprises adjusting the transfer function to effect a relative change (e.g., reduce volume). In some cases, reprogramming the signal processor comprises implementing a predefined transfer function in response to received data, such as location data indicating the wearer is in a particular location. In some such examples, a plurality of pre-programmed transfer functions are stored in a memory and can be implemented based on data acquired via one or more sensors of the external device.

(136) In some embodiments, the external device can be configured to provide an input signal based on audio generated by the external device. For example, the external device can be a smartphone, and can provide an input signal to a wearers implantable battery and/or communication module comprising audio from a phone call, text to speech audio (e.g. reading a text message or an article out loud), and/or media audio (e.g. videos, music, games, etc.). The implantable battery and/or communication module can be configured to relay the input signal to the signal processor for the signal processor to convert into corresponding stimulation signals.

(137) As discussed herein, various devices may be paired to a cochlear implant system. For example, FIG. 9 provides a variety of external devices such as a programmer **900**, charger **902**, smartphone/tablet **904**, Smartwatch/Wearable **906**, Fob **908**, or the like. Additionally or alternatively, external devices may comprise a remote, a remote microphone, an external device that connects to a TV or other AV components for streaming audio to a cochlear implant, or the like. Several examples of external devices are described herein.

(138) In some embodiments, a user (e.g. a physician, audiologist, or the like) may manually pair various external devices with an implantable cochlear implant system. However when pairing multiple external devices, the process may become burdensome for the user.

(139) FIG. 10 shows an illustration of an example pairing system that can be used to facilitate pairing one or more external devices with an implantable cochlear implant system. Such a pairing system can be used to place multiple external devices into communication with a programming device, which can pair each of the external devices with an implantable cochlear implant system.

(140) As shown in the example of FIG. 10, the pairing system **1000** comprises an external pairing device **1060** which can be used to facilitate the pairing of one or more external devices **1036A-E** with a cochlear implant system, such as the fully implantable cochlear implant systems shown in FIGS. 1 and 4. The exemplary pairing device may further include a plurality of near field

communication devices **1066A-E**. Near field communication devices **1066A-E** can include, for example, coils that can be configured to communicate and provide electrical power wirelessly. (141) In some examples, external pairing device **1060** can include a mat, a tabletop, a box, or the like which can provide a relatively flat surface upon which the one or more external devices **1036A-E** can be placed, such as the top or side surface of the external pairing device. In some embodiments, each of the near field communication devices **1066A-E** may be located beneath such a surface such that one or more external devices positioned on the surface may be positioned proximate a near field communication device and communicate therewith.

(142) Near field communication devices **1066A-E** can be configured to provide communication between a programming device **1072** and the one or more external devices **1036A-E**. Programming device **1072** can be configured to communicate with the one or more external devices **1036A-E**, for example, to enable communication between such external device(s) and an implanted cochlear implant system as described elsewhere herein. In some embodiments, programming device **1072** can include a computer, tablet, smartphone, or other device. In the illustrated example, external pairing device **1060** is in communication with programming device **1072** via a wired connection **1082**. As shown, in the example of FIG. **10**, each near field communication device **1066A-E** has a lead extending from the device to wired connection **1082**. In some examples, the wired connection **1082** from the external pairing device is configured to provide parallel communication between each of the near field communication devices and the programming device **1072**. For example, in some embodiments, a programming device can be in communication with each of a plurality of external devices (e.g., **1036A-E**) simultaneously. In other examples, external pairing device **1060** includes an electronics module configured to multiplex communications between each of the near field communication devices **1066A-E** and the programming device **1072**. In some such embodiments the programming device **1072** can communicate with each of the near field communication devices **1066A-E** individually and can communicate with a plurality of near field communication devices sequentially. Additionally or alternatively, in some examples, external pairing device **1060** can be configured to communicate wirelessly with programming device **1072**, such as via a Bluetooth or other wireless communication.

(143) In some embodiments, communication between the programming device **1072** and the one or more external devices **1036A-E** is established when the one or more external devices **1036A-E** are located adjacent to a corresponding near field communication device **1066A-E**.

(144) In some embodiments, an external housing **1030** may be used to house the one or more external devices **1036A-E** and assist in placing the one or more external devices **1036A-E** adjacent to the one or more corresponding near field communication devices **1066A-E**. In some embodiments, the external housing **1030** may comprise any object which can house one or more external devices **1036A-E**, such as a box, a briefcase, various containers, or the like.

(145) In some embodiments, the external housing **1030** may comprise one or more compartments **1034A-E**. As illustrated in FIG. **10**, each of the one or more compartments **1034A-E** may comprise a similar geometrical shape. Alternatively, the one or more compartments may comprise a variety of geometrical shapes. In some embodiments, one or more of the one or more compartments **1034A-E** may be shaped to receive a unique corresponding external device. In such embodiments, the compartments may comprise indicia, such as labels, notifying a user which unique external device should be placed in said compartment.

(146) In some embodiments, the external housing **1030** may comprise at least one relatively flat surface proximate the one or more compartments **1034A-E** that can be placed adjacent to a similarly flat surface of the external pairing device **1060**. For instance, in some examples, the external housing **1030** includes a first surface **1032** configured to engage a second surface **1062** included on the external pairing device **1060**.

(147) As shown in FIG. **10**, the compartments **1034A-E** may be arranged in a pattern to facilitate alignment of one or more external devices (e.g. external devices **1036A-E**) with a corresponding

one or more near field communication devices (e.g. near field communication devices **1066A-E** of the external pairing device **1060**). Such alignment can result in each of the one or more external devices **1036A-E** being positioned near enough to a corresponding one of the near field communication devices **1066A-E** to establish communication with the corresponding near field communication device. Established communication between an external device (e.g., **1036A**) and a corresponding near field communication device (e.g., **1066A**) can result in established communication between the external device (e.g., **1036A**) and the programming device **1072** via the near field communication device (e.g., **1066A**).

(148) In some embodiments, each of the one or more compartments **1034A-E** may be arranged in a first configuration such that each of the one or more compartments **1034A-E** has a unique position within the external housing relative to the first surface **1032**. In some examples, each of the one or more near field communication devices **1066A-E** is arranged in a corresponding configuration relative to the second surface **1062** of the external pairing device **1060**. The corresponding configuration can be such that each of the one or more near field communication devices **1066A-E** of the external pairing device **1060** can be simultaneously aligned with a corresponding compartment **1034A-E** of the external housing **1030**. Similarly, each of one or more external devices **1036A-E** within a corresponding compartment **1034A-E** can simultaneously align with a corresponding one of the one or more near field communication devices **1066A-E**.

(149) Additionally or alternatively, in some embodiments, indicia or other markings may be used to assist in providing a correct alignment between the one or more compartments **1034A-E** and corresponding near field communication devices **1066A-E**. For example, indicia **1068** can be present on the second surface **1062** of the external pairing device **1060** to represent a location and an orientation for positioning the external housing **1030** for correct alignment. Other markings may include indicia on the external housing **1030**, bumps or indentations to provide information on a location and orientation for positioning the external housing **1030**, or the like.

(150) In addition to facilitate pairing, the external pairing device **1060** may also be configured to electronically charge one or more external devices **1036A-E**. For example, the external pairing device **1060** can provide electrical power to one or more external devices **1036A-E** via corresponding near field communication devices **1066A-E** when an external device **1036A-E** is aligned with a corresponding near field communication device **1066A-E**. As described elsewhere herein, a near field communication device (e.g., **1066A**) can comprise a coil configured to facilitate communication with and charging of a corresponding external device (e.g., **1036A**). In various embodiments, electrical power can be provided via the programming device (e.g., via USB connection) and/or an external power source, such as from a wall receptacle, USB port, or the like. In some examples, each near field communication device can include a corresponding drive circuit configured to provide an AC signal to the near field communication device even if receiving power from a DC power source.

(151) In some examples, every near field communication device is able to provide electrical power to charge a corresponding external device simultaneously regardless of which or how many external devices are in communication with the programming device **1072**. Some such examples include a connection to a separate power source, such as a wall outlet. In other examples, only those devices in communication with the programming device **1072** receive electrical power, for example, from the programming device **1072** itself.

(152) While generally described herein with respect to cochlear implant systems (e.g., fully implantable cochlear implant systems), it will be appreciated that configurations such as those shown in FIG. **10** can be used in a variety of applications. In general, an external pairing device including a plurality of near field communication devices can be used to charge and/or communicate with a plurality of devices within an external housing arranged in a configuration corresponding to the configuration of the plurality of near field communication devices. Such systems can be used, for example, to pair each of the plurality of devices to a wireless device. This

can be useful when the wireless device does not have an interface via which a user can initiate pairing with the wireless device itself and/or when a plurality of devices are to be paired with the wireless device. Such systems can be implemented, for example, in a medical device context, such as when one or more external devices are to be paired (e.g., placed into wireless communication) with an implanted medical device.

(153) FIG. **11** shows an example illustration of how a programming device and external pairing device can be used to establish a connection between an external device and an implanted cochlear implant system. For simplicity, FIG. **11** illustrates a single external device **1136**, however, in various embodiments, a plurality of external devices and corresponding near field communication devices can be used, for example, as described with respect to FIG. **10** and elsewhere herein. As shown in FIG. **11**, external device **1136** may establish communication with a near field communication device **1166** of an external pairing device **1160** via a first wireless communication link **1181a**, for example, as discussed with respect to FIG. **10** and elsewhere herein. In some embodiments, near field communication device **1166** can include a coil configured to communicate wirelessly with a corresponding coil in the external device **1136**.

(154) In some embodiments, the near field communication device **1166** can be connected to a programming device (e.g. programming device **1172**), shown in FIG. **11** via communication link **1181b**. While shown as being a wired connection in FIG. **11**, in various embodiments, communication link **1181b** can include a wired or wireless communication link. In some embodiments, the programming device **1172** may be in communication with the external device **1136** through external pairing device **1160** and near field communication device **1166** (e.g., via communication link **1181b** and wireless communication link **1181a**).

(155) Additionally or alternatively, in some embodiments, the programming device **1172** may communicate directly with external device **1136**, such as via a wireless communication link. In such embodiments, external pairing device **1160** may help facilitate initiating the communication between external device **1136** and programming device **1172**. For example, in some embodiments, near field communication between external device **1136** and programming device **1172** (via the external pairing device **1160** and wireless communication link **1181a**) can be used to enable additional wireless communication (e.g., Bluetooth communication) between the programming device **1172** and the external device **1136**, for example, as described in U.S. patent application Ser. No. 16/797,396, filed Feb. 21, 2020, which is assigned to the assignee of the instant application and is incorporated by reference.

(156) In some embodiments, once communication is established between the programming device **1172** and the external device **1136** (e.g. via near field communication device **1166** of external pairing device **1160**), the programming device **1172** may be configured to enable communication between the external device **1136** and an implanted cochlear implant system **1180** such as those systems described herein (e.g., a fully implantable cochlear implant system such as shown in FIGS. **1** and **4**).

(157) Enabling communication between the external device **1136** and the implanted cochlear implant system **1180** may comprise providing information to the external device **1136** to allow communication between the external device **1136** and the implanted cochlear implant system **1180** to be established, such as via communication link **1182**. In some embodiments, the programming device **1172** can be configured to communicate information to the external device **1136** to facilitate wireless communication (e.g., Bluetooth communication) between the external device **1136** and one or more components of the implanted cochlear implant system **1180** (e.g., an implantable battery and/or communication module **1140**).

(158) As shown in FIG. **11**, the implanted cochlear implant system **1180** comprises an implantable battery and/or communication module **1140** configured to communicate with external sources (e.g. external device **1136**, programming device **1172**, or the like). In some embodiments, the implantable battery and/or communication module **1140** may be configured to send and/or receive

wireless communication signals, such as via communication links **1182** and/or **1183**. In various examples, wireless communication links **1182** and **1183** may comprise a wireless connection such as a Bluetooth connection, a Wi-Fi connection, or the like. Accordingly, in some such examples, information provided to the external device **1136** to enable communication between the external device **1136** and the implanted cochlear implant system **1180** can be information enabling Bluetooth or other wireless communication between an implantable battery and/or communication module **1140** and the external device **1136**.

(159) In some examples, programming device **1172** can communicate a password or other key to the external device **1136** to enable communication with implanted cochlear implant system **1180**. Additionally or alternatively, programming device **1172** can communicate identification information to the external device **1136** identifying the implanted cochlear implant system **1180** for which communication is being enabled. Such identification information can include, for example, a media access controller (MAC) address for the implanted cochlear implant system **1180**.

(160) Additionally or alternatively, enabling communication between the external device **1136** and the implanted cochlear implant system **1180** may comprise providing information to the implanted cochlear implant system **1180**, such as via communication link **1183**, to allow communication between the implanted cochlear implant system **1180** and the external device **1136**.

(161) In some embodiments, the communicating information from the programming device **1172** to the external device **1136** to enable communication with the implanted cochlear implant system **1180** is performed in response to a user input to the programming device. For instance, in some examples, programming device **1172** includes a user interface via which a user can input information corresponding to the implanted cochlear implant system **1180** for which communication is to be enabled. In some such embodiments, the programming device **1172** includes a database of identifying information by which a user can select the implanted cochlear implant system **1180** for which to enable communication from the database. For example, the programming device **1172** can include fitting software with information associating an implanted cochlear implant system **1180** to its wearer, and a user (e.g., an audiologist) can enable communication with the implanted cochlear implant system **1180** by selecting the wearer's system within such a database.

(162) Additionally or alternatively, in some examples, the programming device **1172** can be configured to receive information from the implanted cochlear implant system **1180** (e.g., via communication link **1183**). Such information can be used by the programming device to communicate the necessary information to the external device **1136** to enable communication with the implanted cochlear implant system **1180**.

(163) In some embodiments, once information is provided (e.g., via programming device **1172**) to enable communication between the external device **1136** and the implanted cochlear implant system **1180**, information enabling communication is saved in a memory (e.g., in the external device **1136** and/or the implanted cochlear implant system **1180**). In such examples, subsequent communications between the external device **1136** and the implanted cochlear implant system **1180** can be established without requiring external pairing device **1160** or programming device **1172**. Alternatively, subsequent communications between the external device **1136** and the implanted cochlear implant system **1180** may need to be re-enabled by the external pairing device or a similar device.

(164) FIG. **12** illustrates an exemplary method for establishing a connection between an external device and an implantable cochlear implant system as discussed herein. Initially, the one or more external devices (e.g. external devices **1036A-E**) may be placed proximate an external pairing device (e.g. external pairing device **1060**). In some embodiments, as shown in step **1210**, such a step may comprise positioning an external housing (e.g. external housing **1030**) comprising the one or more external devices in compartments (e.g. compartments **1034A-E**) proximate an external pairing device (e.g. external pairing device **1060**). The method further includes establishing

communication between the one or more external devices (e.g. external devices **1036A-E**) and the external pairing device, such as via the one or more near field communication devices (e.g. near field communication devices **1066A-E**) (**1220**). As discussed herein, establishing communication between external devices and the external pairing device can include aligning each of the one or more external devices with a corresponding near field communication device of the external pairing device.

(165) After communication is established, the method further includes communicating information from a programming device (e.g., **1172**) to the external device(s) via the external pairing device (**1230**). For instance, in some embodiments, information may be communicated to the one or more external devices using a connection between the one or more external devices and the external pairing system, such as via the corresponding near field communication device. Additionally or alternatively, information may be communicated using other means, such as directly from the programming device to the implanted system.

(166) The method shown in FIG. **12** further includes enabling communication between one or more external devices and the implant system (**1240**). In some embodiments, the information communicated in step **1230** may comprise information to enable communication between the one or more external devices and the implantable cochlear implant system as described elsewhere herein.

(167) As described, in some examples, an external pairing device (e.g., embodied as a mat with one or more coils embedded therein) can provide communication between a programming device and one or more external devices. In some examples, one or more devices paired to an implanted system can be used to subsequently pair additional devices, for example, as described in U.S. patent application Ser. No. 16/797,396, which is incorporated by reference.

(168) FIG. **13** shows a process flow diagram showing an exemplary method for pairing another device with an implanted system using a paired device, in this case, a charger. The method includes selecting an option to pair a device to an implant on the charger (step **1300**), turning on the desired device and placing it in a pairing mode (step **1302**). The implant determines the devices available for pairing and communicates a list of available devices to the charger (step **1304**), which displays the list of available devices to a user (step **1306**). The user can select from a list of displayed devices to initiate the pairing (step **1308**). The charger and/or selected device can determine if the pairing was successful step (step **1310**). If the pairing is successful, a “pair successful” message can be displayed via the charger and/or the newly-paired device (step **1312**). If the pair was unsuccessful, a “pair not successful” message can be displayed on the charger (step **1314**). For example, in some embodiments, after attempting to initiate pairing between an implant (e.g., via the implantable battery and/or communication module of a system) and another device (e.g., step **1308**), if, after a predetermined amount of time, the charger does not receive an indication confirming pairing from either the implant or the selected device, the charger may determine that the pair was unsuccessful, output the “pair not successful” message (step **1314**), and stop attempting to establish the pairing.

(169) In various examples, devices that can be paired to an implant (e.g., for communication with an implantable battery and/or communication module) via the charger such as via the method shown in FIG. **13** can include a remote, a smart device running an application for interfacing with the implant, a fob, an audio streaming device, or other consumer electronics capable of wireless communication (e.g., Bluetooth).

(170) Accordingly, in some examples, an external pairing device and programming device can be used to pair one or more external devices, including a first external device (e.g., a charger), with an implanted system. The first external device can later be used to pair additional external devices to the system such as via the process shown in FIG. **13**.

(171) As described, in various embodiments, different external devices can interface with implanted components to adjust operation of the system in various ways. In some embodiments,

not all components are capable of performing the same functions as other components. FIG. 14 is a chart showing the various parameters that are adjustable by each of a variety of external devices according to some exemplary systems. In the example of FIG. 14, entries in the chart including an 'X' represent a component configured to perform a corresponding function. Other examples are possible in which different components include different functionality than is represented by the example of FIG. 14, for instance, wherein components other than or in addition to the charger can initiate wireless pairing with the implanted system.

(172) In the illustrated example of FIG. 14, some components can be configured to broadcast an audio stream to the implanted system, such as via a Bluetooth connection. For instance, in some embodiments, an external audio source can broadcast a wireless signal to an implanted system, such as a signal representative of inputs received into the external audio source (e.g., from another media device). Additionally or alternatively, systems can include a remote audio pickup having a microphone or other audio sensing device to receive sounds and to broadcast a wireless signal representative of the received sounds.

(173) In various examples, one or more devices in the chart of FIG. 14 can be prepackaged in an external housing (e.g., housing 1030 of FIG. 10) for delivery to a wearer. As described elsewhere herein, each such device can be located within a corresponding compartment (e.g., 1034A-E) of the external housing. The external housing can be positioned proximate a pairing device and aligned such that each compartment is proximate a corresponding near field communication device. Each external device within a compartment can be placed into communication with a programming device via the corresponding near field communication device. The programming device can be used to pair such external devices with a cochlear implant system, for example, by communicating a MAC address associated with the system to each of the external devices.

(174) In some embodiments, a user may select a prepackaged set of external devices, for example, from a list of available packages or by creating a custom set of external devices. Such devices can be packaged into an external housing by an audiologist or delivered to an audiologist as a prepacked set of external devices. The audiologist can then position the external housing on the external pairing device and use the external pairing device and programming device to pair each external device in the external housing to a wearer's implanted cochlear implant system without having to remove any device from its packaging. Moreover, as described herein, the near field communication devices of the external pairing device can be used to provide electrical power to and charge the external devices. Accordingly, an audiologist can at least partially charge a plurality of external devices and pair such devices to a wearer's implanted cochlear implant system without removing any such devices from the packaging. The audiologist can show the wearer how the external devices work using in-office models while the wearer's external devices remain in their packaging.

(175) As described herein, in some embodiments, a first surface (e.g., 1032) of an external housing (e.g., 1030) and a corresponding second surface (e.g., 1062) of an external pairing device (e.g., 1060) can both be flat. In some such examples, the flat first surface of the external housing can be placed on the second flat surface of the external pairing device to enable communication between one or more external devices and a programming device as discussed herein. Similarly, in some examples, the external pairing device can be placed on top of the external housing to similarly establish communication between one or more external devices and a programming device.

(176) In some examples, the first surface and second surface need not be flat. For instance, in some embodiments, at least one of the first surface 1032 and the second surface 1062 can comprise various textures and/or geometrical features, such as angled or curved surfaces. In some such examples, the first and second surface can include complementary shapes such that the compartments arranged in unique positions relative to the first surface are each positioned proximate a corresponding near field communication device in the second surface when the first and second surfaces are aligned.

(177) In various examples, the first and second surface can be generally horizontal surfaces, such as a mat comprising the second surface and one or more coils embedded therein acting as near field communication devices and configured to support a box comprising one or more compartments and the first surface. In other examples, the first and second surface need not be configured horizontally. In an example embodiment, the second surface can be an approximately vertical surface, and the external housing can be placed adjacent the second surface. In some such examples, the external housing can be set down on a surface other than the first surface such that the first surface on the external housing is positioned proximate the second surface. Other configurations are possible and within the scope of this disclosure.

(178) It will be appreciated that, while generally described herein with respect to implantable hearing systems, communication techniques described can be used in a variety of other implantable systems, such as various neuromodulation devices/systems, including, for example, pain management, spinal cord stimulation, brain stimulation (e.g., deep brain stimulation), and the like. Techniques described herein can be used for pairing external devices to a variety of types of systems, such as partially and fully implantable systems.

(179) Various non-limiting embodiments have been described. These and others are within the scope of the following claims.

Claims

1. A method comprising: receiving an external housing carrying a plurality of external devices in a corresponding plurality of compartments at an external pairing device such that each of the plurality of external devices is positioned proximate a corresponding near field communication device of the external pairing device; establishing communication between a programming device and each of the plurality of external devices using the external pairing device; and simultaneously charging each of the plurality of external devices carried by the external housing using the external pairing device.
2. The method of claim 1, further comprising: communicating information from the programming device to each of the plurality of external devices to enable communication between each of the plurality of external devices and a cochlear implant system.
3. The method of claim 2, wherein communicating information from the programming device to each of the plurality of external devices to enable communication between each of the plurality of external devices and the cochlear implant system comprises communicating a media access control (MAC) address associated with the cochlear implant system to each of the plurality of external devices.
4. The method of claim 2, wherein communicating information from the programming device to each of the plurality of external devices comprises communicating information to each of the plurality of external devices individually and sequentially.
5. The method of claim 2, wherein the programming device comprises a computer including fitting software associating the cochlear implant system with a wearer.
6. The method of claim 5, wherein the fitting software has access to a database of cochlear implant systems, and wherein enabling communication between the plurality of external devices and the cochlear implant system comprises receiving a selection of the cochlear implant system from within the database of cochlear implant systems.
7. A method comprising: placing an external housing carrying a plurality of external devices in a corresponding plurality of compartments at an external pairing device such that each of the plurality of external devices is positioned proximate a corresponding near field communication device of the external pairing device; establishing communication between a programming device and each of the plurality of external devices using the external pairing device; and simultaneously charging each of the plurality of external devices carried by the external housing using the external

pairing device.

8. The method of claim 7, further comprising packaging each of the plurality of external devices into one of the plurality of compartments.

9. The method of claim 7, further comprising selecting a prepackaged set of external devices, the external housing carrying the plurality of external devices corresponding to the selected prepackaged set of external devices.

10. The method of claim 7, wherein placing the external housing carrying the plurality of external devices in the corresponding plurality of compartments at the external pairing device such that each of the plurality of external devices is positioned proximate a corresponding near field communication device of the external pairing device comprises placing a first surface of the external housing onto a second surface of the external pairing device.

11. The method of claim 7, further comprising establishing wireless communication between a cochlear implant system and the programming device.

12. The method of claim 7, wherein charging each of the plurality of external devices comprises providing power to each of the plurality of external devices simultaneously using the programming device.

13. The method of claim 7, wherein charging each of the plurality of external devices comprises providing power to each of the plurality of external devices simultaneously comprises using a wall outlet or USB.

14. The method of claim 7, wherein the external housing comprises a first surface and the external pairing device comprises a second surface, the second surface of the external pairing device comprising indicia representing a desired location and a desired orientation of the external housing, and wherein placing the first surface of the external housing onto the second surface of the external pairing device comprises positioning the external housing relative to the indicia.

15. A cochlear implant communication system comprising: a plurality of external devices, each of the plurality of external devices being capable of wireless communication with a cochlear implant system; an external housing having: a first surface; and a plurality of compartments arranged in a first configuration, each of the plurality of compartments having a unique position within the external housing relative to the first surface, wherein each of the plurality of external devices is positioned in a different one of the plurality of compartments of the external housing; and an external pairing device comprising: a second surface; and a plurality of near field communication devices arranged in a configuration corresponding to the first configuration relative to the second surface such that the first surface of the external housing can be aligned with the second surface of the external pairing device in such a way that each of the plurality of near field communication devices aligns with a corresponding one of the plurality of compartments of the external housing; and wherein the external pairing device is configured to facilitate simultaneous charging of each of the plurality of external devices using a corresponding one of the plurality of near field communication devices when the first surface of the external housing engages the second surface of the external pairing device.

16. The cochlear implant communication system of claim 15, wherein the external pairing device system is configured to connect to a programming device and provide communication between said programming device and each of the plurality of external devices.

17. The cochlear implant communication system of claim 15, further comprising a programming device in communication with the external pairing device and configured to enable communication between each of the plurality of external devices and the cochlear implant system.

18. The cochlear implant communication system of claim 17, wherein the programming device is configured to communicate a media access control (MAC) address associated with the cochlear implant system to each of the plurality of external devices.

19. The cochlear implant communication system of claim 15, wherein the external pairing device is configured to connect to a power source.

20. The cochlear implant communication system of claim 19, wherein the power source is an electrical outlet or a USB power source.
